Memory-Augmented Detection Transformer for Few-Shot Object Detection in Remote Sensing Imagery

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***Abstract*— The few-shot object detection (FSOD) in remote sensing (RS) faces a fundamental challenge in balancing contextual feature learning with the representation stability. Current approaches either excel at capturing rich contextual relationships through multimodal architectures but struggle with catastrophic forgetting or maintain stable feature representations through incremental learning frameworks at the cost of contextual understanding. This limitation is particularly acute in RS, where complex spatial–contextual dependencies form distinctive co-occurrence patterns important for accurate object detection. We present a memory-augmented detection transformer (MemDeT) that bridges this gap through three key modules: 1) a contextual layerwise fusion (CLF) module that progressively integrates visual and textual information across transformer layers through an adaptive attention mechanism and unified cross-modal fusion, enabling both fine-grained object feature extraction at lower layers and abstract contextual relationship learning at higher layers; 2) a unified episodic memory (UEM) that serves as a dynamic knowledge repository, employing similarity-surprise-based update mechanisms with structured key–value memory organization to strategically retrieve and update relevant past experiences while preserving base knowledge through contextual scoring and dynamic retrieval; and 3) a memory-augmented decoder (MAD) that generates context-aware queries by combining current visual observations with accumulated contextual knowledge through memory-prioritized attention and progressive query refinement. Extensive experiments on RS datasets, NWPU VHR-10, DIOR, and iSAID, demonstrate that MemDeT significantly outperforms state-of-the-art methods in both contextual understanding and knowledge retention. Cross-dataset evaluation further validates MemDeT's robust generalization capabilities, successfully transferring knowledge from DIOR to iSAID's urban scenes, particularly in scenarios requiring strong contextual understanding with limited training examples.**

***Index Terms*— Catastrophic forgetting, few-shot object detection (FSOD), image–text alignment, memory-augmented detection, remote sensing (RS).**

I. Introduction

THE analysis of remote sensing images (RSIs) has inspired research across several domains, including image classification [1], [2], semantic segmentation [3], [4], and object detection [5], [6]. Among these, the object detection has broader applications, such as environmental monitoring [7], [8] and agricultural management [9], [10]. While state-of-the-art remote sensing (RS) detectors [11], [12], [13], [14] achieve impressive results, they require extensive labeled data for training, which is labor intensive to annotate. This, combined with the inherent class imbalance in real-world data, underscores the pressing need for methods that can learn effectively from limited examples.

Few-shot object detection (FSOD) has emerged as a promising paradigm to address this challenge, aiming to detect objects using only a few labeled examples. Current methods for FSOD in RS suffer from a critical tension between contextual feature learning and representation stability, manifesting in the adaptation of feature space representations. This challenge is particularly evident in how current approaches manage their feature spaces, developing along several directions with different tradeoffs. Traditional unimodal approaches [15], [16], [17], [18], [19] rely solely on visual features for detection, utilizing techniques like feature-level adjustments and pretrained model adaptation. While these methods provide a foundation for FSOD, they struggle to capture complex contextual relationships due to their reliance on single-modality features. Multimodal architectures [20], [21], [22] attempt to overcome these limitations by integrating visual and textual information. These methods construct comprehensive feature spaces through cross-modal integration, where visual features are enhanced with semantic knowledge from textual descriptions. This integration enables the capture of complex contextual relationships and semantic dependencies between objects, enhancing the model's ability to understand scene context beyond what is possible with visual features alone. However, when adapting to novel categories, these rich feature spaces become unstable. The process of incorporating new classes disturbs the carefully constructed cross-modal relationships for base categories, leading to catastrophic forgetting. This instability manifests as significant performance

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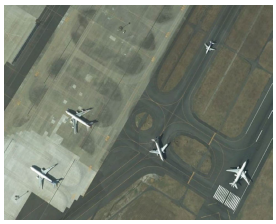
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The planes are arranged along the runways and the taxiways.



A densely populated urban area with a major highway running through it.

Fig. 1. Utilizing contextual information in FSOD strengthens the memory embeddings within the UEM, leading to enhanced knowledge retention. (Left) Planes positioned along runways and taxiways provide spatial cues that reinforce the MemDeT's memory of airport-specific contexts. (Right) Densely populated urban area with a highway, anchoring knowledge of urban layouts and infrastructure. By embedding both visual and textual context, the model builds more robust and distinctive memory representations, allowing it to retain knowledge more effectively and minimize forgetting over time.

degradation on base classes, with accuracy often dropping substantially after adaptation to new categories. In contrast, incremental learning frameworks [23], [24], [25] prioritize maintaining stable feature representations for base categories. These approaches implement strict constraints on feature space modifications during the adaptation process, effectively preserving knowledge of base classes. They achieve this through various mechanisms, such as parameter regularization, feature space constraints, and prototypical learning, ensuring that updates for new categories minimally impact the existing feature representations. However, this conservative approach comes at a significant cost. The rigid constraints on feature space modifications severely limit the model's ability to learn and represent new contextual relationships. This results in a rigid representation space that, while stable for base categories, fails to capture the rich contextual relationships necessary for effective few-shot detection.

These limitations are particularly challenging in RS applications due to their unique characteristics. RS scenes feature complex spatial–contextual dependencies that form distinctive co-occurrence patterns. For instance, the airport scene, as shown in Fig. 1, demonstrates how planes are systematically arranged along runways and taxiways, while the urban scene shows how densely populated areas typically develop along major highways. These intricate spatial relationships are important for accurate interpretation, detecting a runway without considering its relationship to aircraft, or identifying urban development patterns without understanding road network connections, would miss essential context. Learning and preserving such complex dependencies pose a significant challenge. When models adapt to new object categories, they often forget these crucial spatial relationships for previously learned classes; a runway might still be detected, but its learned relationship with aircraft may be lost. This problem of catastrophic forgetting is further compounded by the limited availability of annotated data for rare or emergent object categories, making it essential to develop approaches that can learn from a few examples while maintaining previously learned contextual relationships. In this context, existing methods struggle to capture such meaningful dependencies between contextually related objects, despite their ability to detect individual objects independently.

This motivates our development of the first memory-augmented detection transformer (MemDeT) for FSOD in RS that reimagines how knowledge can be preserved and adapted. At its core, MemDeT introduces three innovative components to address the challenges of contextual feature learning and representation stability. The contextual layerwise fusion (CLF) module aligns visual and textual information across multiple transformer layers. CLF performs iterative refinement at each transformer layer through a progressive integration mechanism. This layerwise processing enables the model to capture both fine-grained object features at lower layers and abstract contextual relationships at higher layers. The unified episodic memory (UEM) serves as a dynamic knowledge repository that evolves with the model's learning experience. UEM maintains and continuously updates a diverse set of episodic memories that capture complex interactions among object appearances, contextual relationships, and spatial configurations. When the model encounters new object categories, UEM employs a strategic retrieval mechanism that selects relevant past experiences based on both visual and contextual similarity. These retrieved memories are then adaptively integrated with current observations, enabling the model to leverage previously learned patterns while maintaining the flexibility to adapt to novel scenarios. This dynamic memory management ensures that the model can preserve base knowledge while effectively generalizing to new classes. Building upon these capabilities, MemDeT incorporates a memory-augmented decoder (MAD) that enhances how object detection queries are generated and processed. MAD introduces a memory-prioritized query generation mechanism that combines current visual observations with contextually relevant episodic memories. The decoder constructs rich, context-aware query vectors by integrating immediate visual evidence with accumulated contextual knowledge from the memory bank, as illustrated in Fig. 2. In this article, we present the following contributions to advance FSOD in RS.

1) We present MemDeT, the first memory-augmented detection transformer for FSOD in RS. Our approach uniquely addresses the critical tension between contextual features and representation stability in FSOD by integrating dynamic memory mechanisms with multimodal feature learning, enabling effective adaptation to novel categories while maintaining rich contextual understanding.
2) We propose a CLF module that progressively aligns visual and textual information across transformer layers, enabling both fine-grained feature extraction and high-level contextual relationship modeling.
3) We introduce UEM that dynamically evolves with model learning, maintaining and updating diverse memories of object appearances and contextual relationships while preserving base-class knowledge.
4) We develop MAD with novel query generation that combines current observations with episodic memories,

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TABLE I
COMPARISON OF MEMDET PROPOSED MODULES WITH EXISTING ARCHITECTURAL APPROACHES FOR FSOD

Aspect	Existing Methods	MemDeT
Multi-modal Feature Learning	- Direct visual-text alignment [21] - Single-stage feature fusion [22] - Limited contextual understanding [20]	- Progressive layer-wise fusion - Hierarchical contextual building - Multi-level semantic integration
Knowledge Preservation	- Fixed parameter constraints [24] - Separate base/novel features [23] - Static feature representations [26]	- Dynamic memory organization - Unified feature space - Selective memory updates
Detection Strategy	- Fixed detection patterns [27] - Limited feature reuse [28] - Context-independent queries [29]	- Memory-guided detection - Historical knowledge integration - Context-aware query generation

ensuring detection benefits from both present visual evidence and past contextual knowledge.

II. RELATED WORKS

A. Object Detection

Object detection has evolved through diverse architectural approaches. Traditional methods used engineered features and filters [30], [31], but often struggled with complex scenes and parameter sensitivity. Deep learning-based methods have significantly advanced object detection capabilities. Faster R-CNN [11] advances region proposal generation and multiscale feature extraction, though with substantial computational overhead. FPN [12] builds upon this by introducing feature pyramid networks for multiscale detection but inherits similar computational constraints. YOLO [32] addresses computational efficiency through single-stage detection while struggling with orientation variations, while CenterNet [33] improves detection accuracy through keypoint-based object localization. KCDNet [5] addresses the challenge of inconsistent sensor data in multimodal object detection by employing a knowledge-consistent distillation framework, enabling the model to learn from reliable sensor inputs to compensate for deficiencies in less reliable ones. DETR [13] and its variant deformable DETR [14] eliminate the need for hand-crafted components like anchor boxes but face challenges with arbitrary orientations. STD [34] tackles rotation-sensitive detection through decoupled network branches, while HODet [35] integrates horizontal and oriented detection using Hough transform-based angle estimation. HRDet [36] further improves oriented object detection through hierarchical mask prompting and oriented IoU loss, and CAGT [37] enhances detection in complex scenes through graph-based contextual modeling.

Despite these advancements, these methods share a fundamental limitation in their dependence on extensive annotated data, particularly challenging for domain-specific applications.

B. Few Shot Object Detection

Recent advancements in FSOD have tackled the challenge of learning from limited data through specialized strategies designed for natural images (NIs) and RSIs. In NIs, metalearning approaches in NIs, including RepMet [16] and Meta R-CNN [15], enable rapid adaptation but struggle with feature integration and high-intraclass variance. Fine-tuning methods, such as LSTD [18] and TFA [19], improve feature transferability but face limitations in optimizing for novel classes. FSCE [38] advances this further through contrastive learning, enhancing feature discrimination in limited-data scenarios. FSOD methods in RS face challenges like inconsistent labeling and sample imbalance. FRW [17] uses episodic training with metalearning for limited annotations but lacks transfer learning, reducing representational power. Prototype-CNN [39] enhances features through prototype learning but struggles with new class representativeness. Sample-efficient few-shot object detector (SAE-FSDet) [40] addresses inconsistent labeling but needs sufficient novel instances to generalize effectively. FOMC [41] clusters similar instances for improved class distinction. SMDC-Net [42] uses saliency-guided branches to enhance feature representation while reducing background noise. Understanding negative proposals (UNP) [43] employ gradient relaxation to separate confusing proposals but struggles with inaccurate initial proposals. Yu et al. [44] introduced a selective attention mechanism and pixel-level extractor to improve spatial focus and object positioning. Wu et al. [45] proposed a filtering module that builds class prototypes based on attributes, enhancing classification while managing intraclass variance.

Despite these advancements, a common limitation of these methods is their reliance solely on visual features, which constrains the model's ability to capture the semantic relationships necessary for distinguishing contextual cues, ultimately challenging robust generalization from a limited number of annotated images for new classes.

C. Adaptive Learning Strategies for FSOD

Given the challenges of limited data in FSOD, implementing adaptive learning strategies is essential for enabling models to effectively identify key features when relying solely on visual information. Image–text fusion has proven effective in addressing these challenges by improving the identification of distinguishing features essential for robust generalization. For instance, Bou et al. [46] utilized a prototype-based classifier that leverages embeddings from textual descriptions generated by a pretrained model, allowing for the identification of new classes with minimal examples. Infrs [25] introduced prototypical calibration based on the Wasserstein distance to intelligently adjust the feature space, allowing the model to learn new categories without drastically altering the representations of existing ones. TSF-RGR [22] enhances the

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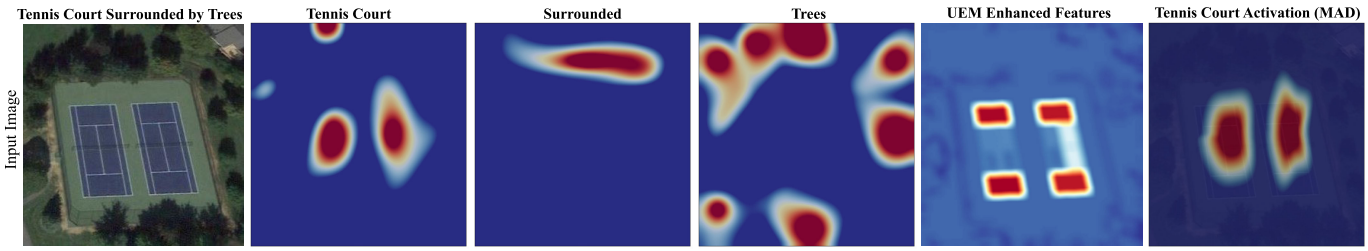


Fig. 2. First column shows the original image with the tennis court and trees. The second, third, and fourth columns display the word-level activation maps from CLF for "tennis court," "surrounded," and "trees," respectively, highlighting their shapes and positions. The fifth column shows UEM-enhanced feature maps, which expand the activations beyond the tennis court, improving recognition with a broader context. The sixth column presents the tennis court activations from the MAD, capturing context-aware, and court-shaped representations.

object detection by creating a structured corpus of attribute descriptions and employing graph structures to capture spatial and semantic relationships among objects. TEMO [21] integrates visual and textual data sources to enrich the feature space, bridging semantic gaps between base and novel classes. In contrast, UMFT [20] introduces an end-to-end multimodal transformer that aligns image and text data without the need for a region proposal network, thereby strengthening the model's capacity to represent object features robustly for new classes. However, these methods effectively enhance model performance but encounter limitations, particularly catastrophic forgetting, which leads to decreased accuracy on previously learned classes as new ones are introduced. Incremental learning methods aim to preserve base knowledge while learning new classes. Current approaches rely heavily on fixed parameter constraints and maintain separate feature spaces for base and novel classes. G-FSDet [24] employs parameter-level constraints through metric-based loss and compensation modules, while FSIOD-DFP [23] uses a frozen base model and task-decoupled detection head. However, these static feature representations and rigid constraints, while preserving base knowledge, significantly limit the model's ability to adapt to novel classes and capture new patterns effectively.

In this work, we propose MemDeT, which adapts to new classes while preserving the performance of previously learned classes by recalling past experiences and effectively capturing complex contextual relationships by incorporating textual descriptions alongside visual features for accurate object detection. Table I shows the contributions made by MemDet compared with existing adaptive learning strategies for FSOD.

D. Memory-Augmented Networks

Episodic memory [47], [48], [49] is crucial for rapid adaptation, facilitating the recall of specific past events and has been effectively applied in various domains, including reinforcement learning and continual learning. In reinforcement learning, Zhu et al. [50] and Gershman and Daw [51] utilized the episodic memory to store past experiences, enhancing the generalization for novel object categories with limited data. In continual learning, gradient episodic memory (GEM) framework [52] utilizes episodic memory to preserve knowledge and mitigate catastrophic forgetting, while Chaudhry et al. [53] enhanced this by introducing experience replay strategies to reinforce representations of previously learned classes. Building upon these, memory-augmented neural networks (MANNs) [54] and matching networks [55] enhance few-shot learning by enabling the classification of new objects through stored exemplars. Metanetworks [56] use a dual-weight system for fast adaptation, while adaptive posterior learning (APL) [57] employs Bayesian updating to preserve prior knowledge, supporting memory retention in dynamic few-shot tasks. Furthermore, the episodic memory optimization (EMO) [58] enhances few-shot learning by retaining gradient histories from past tasks, enabling improved generalization and adaptability to new tasks with limited training samples. Collectively, these studies illustrate the significant role of episodic memory in improving few-shot learning, supporting quicker adaptation, and better generalization in dynamic environments.

Building upon this inspiration, we propose MemDeT, which incorporates UEM as a dynamic repository for storing representations of past experiences. This UEM mechanism enables MemDeT to effectively retrieve relevant instances when new classes are introduced, thereby preventing performance degradation on previously learned classes. This retrieval process not only reinforces the model's understanding of earlier classes but also facilitates adaptation to new features and relationships from novel classes.

III. Methodology

In this work, we propose the MemDeT, as shown in Fig. 3. The process begins with CLF, which receives visual input through ViT and corresponding text through BERT, progressively fuses these features across multiple transformer layers. The lower layers extract fundamental visual patterns (edges, shapes, and textures) and basic semantic concepts from the text, while the middle layers begin aligning these modality-specific features. Higher layers then build complex cross-modal relationships by integrating visual patterns with semantic concepts, creating rich multimodal embeddings that capture both visual and contextual information. These unified embeddings are passed to UEM, which serves as a dynamic knowledge repository. UEM evaluates incoming embeddings against stored memories using similarity and surprise mechanisms. For similar patterns, it retrieves and updates relevant memories, while for novel patterns, it creates new memory entries. This dynamic memory management enables the model to maintain a balance between preserving base knowledge and adapting to new distributions. Finally, MAD leverages both the current CLF-generated embeddings and UEM's retrieved memories to construct memory-prioritized queries. These

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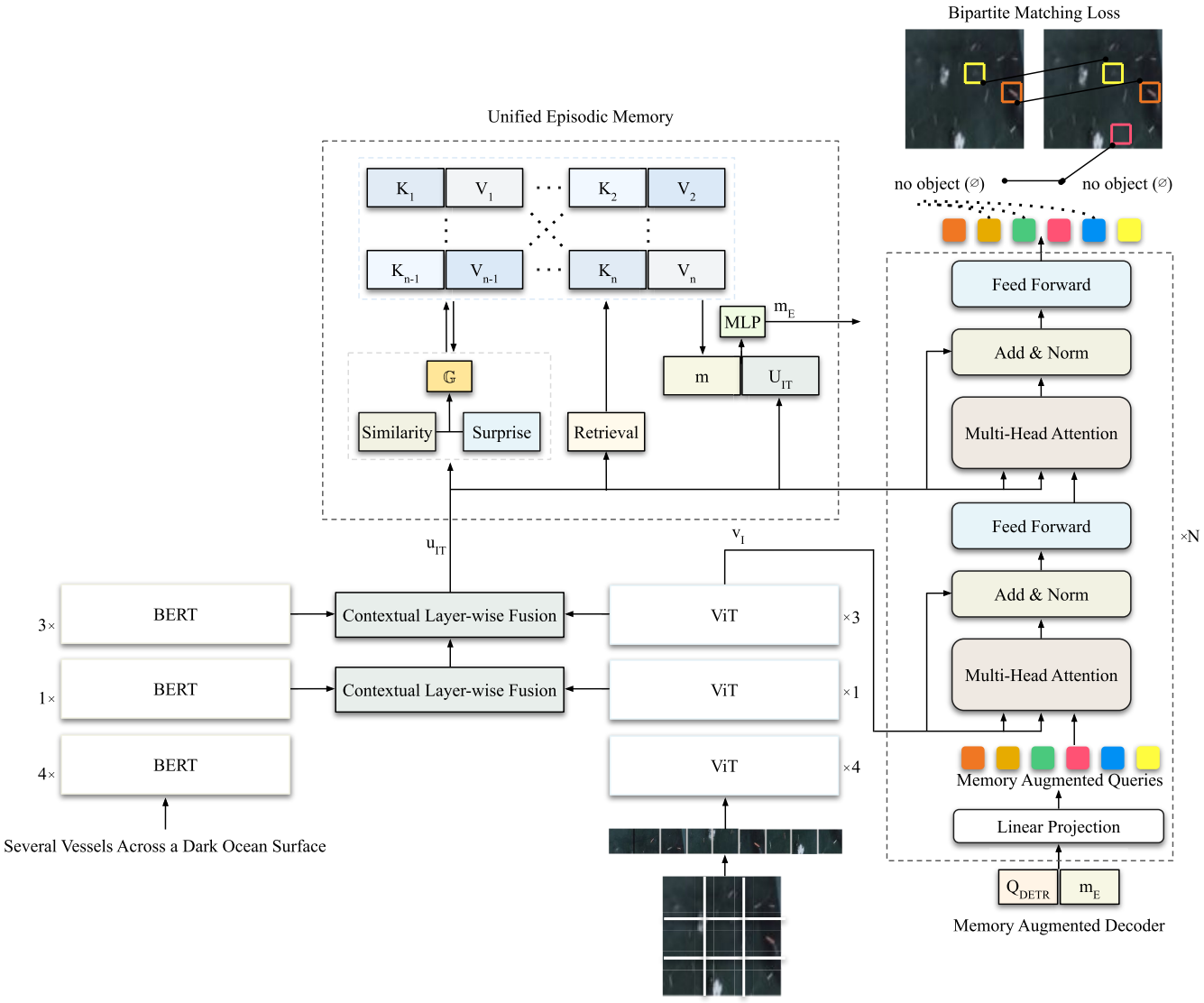


Fig. 3. Architecture of MemDeT. The model integrates three key components: 1) CLF module that aligns visual and textual information across layers to extract details and learn relationships; 2) UEM that stores and updates knowledge based on new experiences; and 3) MAD that creates context-aware queries by combining current visuals with stored knowledge.

queries combine immediate visual–textual evidence with relevant historical context, enabling adaptive object detection that accounts for both current observations and accumulated knowledge. This integrated workflow allows MemDeT to dynamically adjust its detection strategy based on both immediate task characteristics and relevant past experiences. This enables MemDeT to effectively handle both known and novel object classes by maintaining a balance between immediate scene understanding and historical knowledge utilization.

A. Contextual Layer-Wise Fusion (CLF)

We propose CLF to enhance cross-modal integration by processing image and text inputs separately through dedicated encoders before fusion. Specifically, ViT (E_V) [59] takes the image I and outputs a visual embedding $v_I \in \mathbb{R}^{d_v}$, where d_v represents the visual feature dimension. Simultaneously, BERT (E_T) [60] encodes the text T into a textual embedding $t_T \in \mathbb{R}^{d_t}$, with d_t as the textual feature dimension. CLF employs a multilayer fusion strategy, inspired by BridgeTower [61], which merges the visual and textual embeddings at multiple layers rather than at the final layer alone, as shown in Fig. 3. However, CLF improves upon BridgeTower by using a unified cross-modal attention mechanism that selectively emphasizes relevant features from each modality, rather than uniformly fusing them at every layer, as shown in Fig. 4. At each (l^{th}) layer, CLF combines the visual embeddings v_I^l and textual embeddings t_T^l with contextual embeddings from the previous layer u_{IT}^{l-1}, allowing the model to capture diverse feature representations from both high- and low-level concepts. This multilayer approach enables CLF to retain detailed information across layers, enhancing the model's ability to understand intricate interactions between visual and textual elements, as shown in the following equations:

$$\bar{v}_I^l = u_{\text{IT}}^{l-1} + \text{MSA}\left(u_{\text{IT}}^{l-1}, v_I^l, v_I^l\right) \tag{1}$$

$$\bar{t}_T^l = u_{\text{IT}}^{l-1} + \text{MSA}\left(u_{\text{IT}}^{l-1}, t_T^l, t_T^l\right) \tag{2}$$

where MSA represents the multiself-attention.

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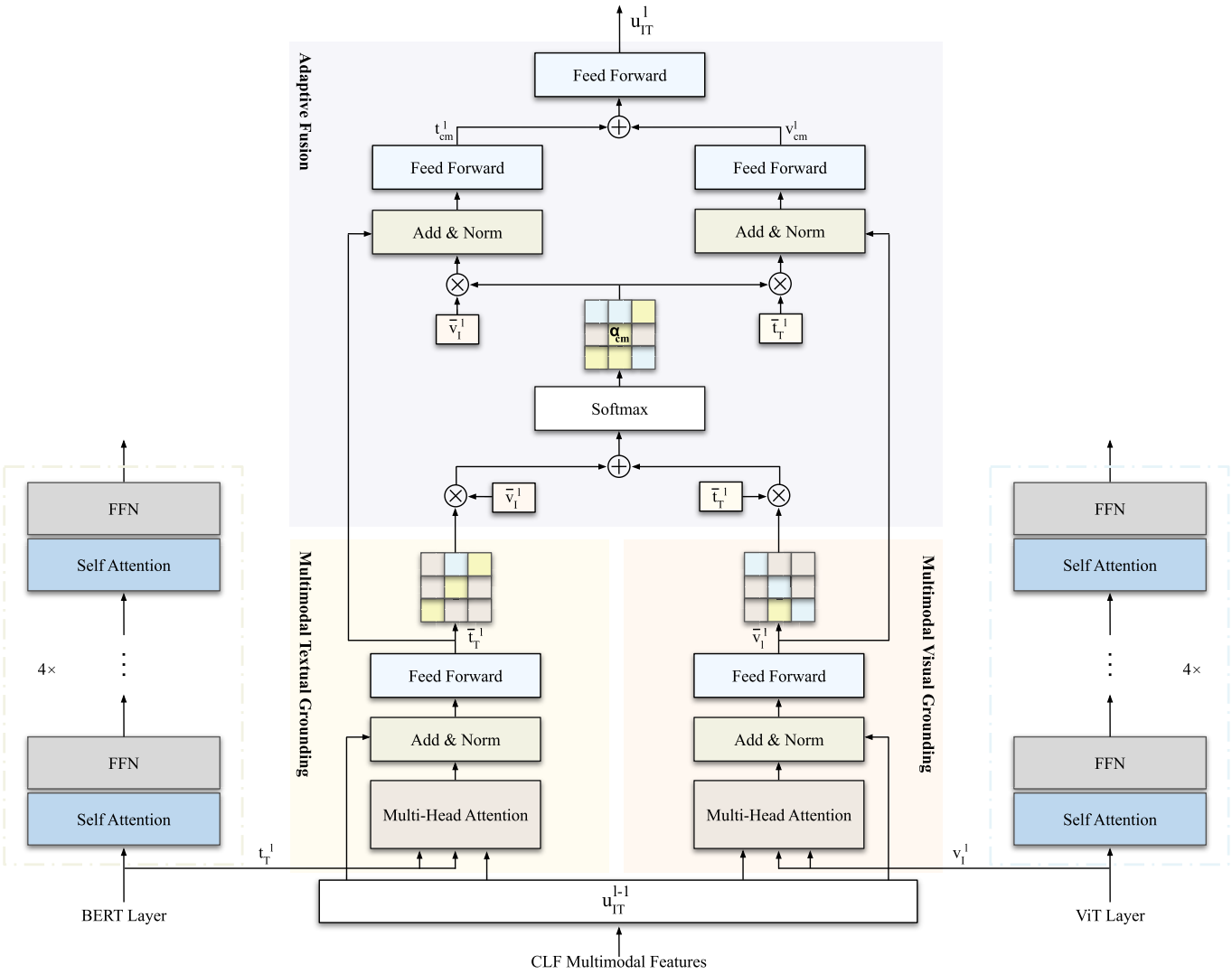


Fig. 4. CLF module processes features in two branches: (left) textual grounding and (right) visual grounding. At each layer l, the BERT (t_i^l) and ViT (v_i^l) features are combined with the previous layer's multimodal context (u_{IT}^{l-1}). This allows each modality to build its own representation while maintaining cross-modal context. (Middle) Adaptive fusion mechanism fuses the features, enabling each modality to focus on relevant aspects of the other, creating a unified representation, where both modalities enhance each other.

To enhance the interaction between visual and textual modalities, we employ a unified cross-modal attention that integrates contextual embeddings from both modalities. Instead of computing attention separately for visual-to-text and text-to-visual, we combine the context-aware embeddings into a single attention computation that refines both representations simultaneously. At each lth layer, we compute the attention weights α_{cm} using (3), which allows us to compute the attended features for both modalities using (4) and (5)

$$\alpha_{\text{cm}} = \text{SM}\left(\frac{\bar{v}_I^l W_Q^l \left(\bar{t}_T^l W_K^l\right)^\top + \bar{t}_T^l W_Q^l \left(\bar{v}_I^l W_K^l\right)^\top}{\sqrt{d}}\right) \tag{3}$$

$$v_{\text{cm}}^l = W_V^l \left(\bar{v}_I^l + \alpha_{\text{cm}} \cdot \bar{t}_T^l\right) \tag{4}$$

$$t_{\text{cm}}^l = W_T^l \left(\bar{t}_T^l + \alpha_{\text{cm}} \cdot \bar{v}_I^l\right) \tag{5}$$

where SM represents softmax and W_V^l and W_T^l are learned projection matrices that align the feature spaces of the respective modalities.

Once $\bar{v}_I^l$ and $\bar{t}_T^l$ are calculated through the unified cross-modal attention, the fused cross-modal embedding u_{IT}^l is computed to capture the combined information from both visual and textual modalities using the following equation:

$$u_{\text{IT}}^l = W_U^l \left(v_{\text{cm}}^l + t_{\text{cm}}^l\right) \tag{6}$$

where $W_U^l \in \mathbb{R}^d$ is a learned projection matrix that maps the combined embeddings into a unified feature space.

After processing through all CLF's L layers, the final multimodal embedding $u_{\text{IT}} = u_{\text{IT}}^l$ represents a richly integrated multimodal embedding that has been refined iteratively, capturing a cohesive representation of the image–text pairs, informed by context-specific interactions across layers.

B. Unified Episodic Memory

We propose UEM (Fig. 5) to address the challenge of knowledge retention in feature representations through a memory-driven mechanism. By constructing a unified memory that stores multimodal representations from past experiences, UEM effectively captures diverse information across modalities, enabling the model to retain important knowledge

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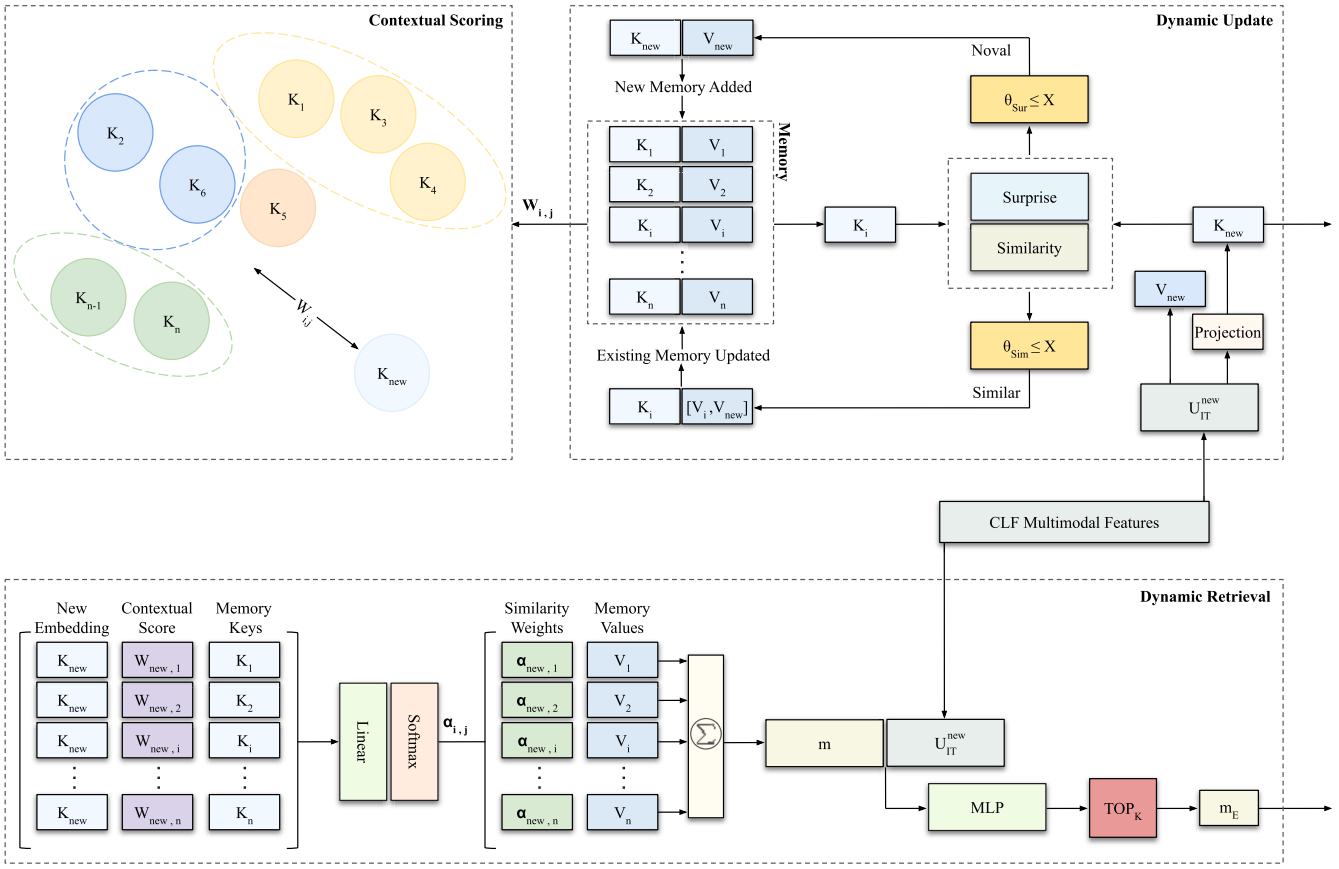


Fig. 5. UEM framework improves knowledge retention by storing past experiences as key–value pairs. The process includes: 1) memory structure, where experiences are stored with keys as compressed embeddings and values containing full-multimodal information; 2) updates, where new experiences are compared with existing memories and either update or add new entries based on similarity and novelty; 3) contextual scoring, which refines memory retrieval by linking stored memories to the current input based on relevance; and 4) retrieval, where relevant memories are selected to help with new tasks, enhancing flexibility and performance.

for future tasks. When encountering a new scenario, UEM first updates the memory bank by adding new information or consolidating similar experiences with existing entries. Then, during retrieval, UEM selects embeddings that share contextual similarity with the current input. This combined update-retrieve process aids in understanding novel classes by providing the model with relevant features and relationships from prior experiences, aligning these with new contexts to support accurate classification and adaptation.

1) Episodic Memory Structure: CLF's multilayer fusion mechanism creates powerful contextual representations by progressively integrating visual and textual information. However, when adapting to new scenarios, previously learned feature alignments (e.g., runway–aircraft spatial relationships) can become unstable, risking the loss of crucial contextual dependencies. To address this, UEM employs a key–value architecture to store unified contextual embeddings (u_{IT}), strategically balancing rapid memory access with comprehensive contextual preservation. Each new experience leads to the construction of compressed key K_i and value V_i for efficient retrieval and utilization of past knowledge, with each memory entry M_i formally represented as (7). This mechanism ensures that task-relevant patterns can be quickly identified through keys while maintaining detailed contextual information through values, preventing interference that could lead to catastrophic forgetting

$$M_i = (K_i, V_i) \tag{7}$$

where $K_i = f_{\mathrm{proj}}(u_{\mathrm{IT}})$ projects the keys into compressed $\mathbb{R}^{d/D}$ space, with $D = 4$ representing the compression ratio. The value $V_i = u_{\mathrm{IT}} \in \mathbb{R}^d$ preserves the complete multimodal embedding for comprehensive information retention.

The episodic memory M comprises multiple entries structured, as shown in (8). This memory structure allows UEM to maintain a rich repository of multimodal knowledge, facilitating effective adaptation and learning when encountering new tasks, thereby enhancing its overall generalization capabilities

$$M = \left[M_1, M_2, \ldots, M_{n-1}, M_n\right]. \tag{8}$$

2) Dynamic Memory Update: Dynamic memory updates require a decision on whether to modify existing memories or create new entries. The continuous modification of existing memory entries can distort stored representations, causing distinct clusters to overlap and lose their discriminative power. On the other hand, creating a new memory entry for every variation fragments the embedding space, making minor shifts create multiple separate clusters, which complicates the model's ability to detect shared patterns. UEM addresses these challenges by employing similarity and novelty mechanisms. When encountering a new as multimodal experience $u_{\mathrm{IT}}^{\mathrm{new}}$,

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it begins by projecting the new embedding to a compressed key representation that preserves essential spatial patterns while enabling efficient comparison

$$K_{\text{new}} = f_{\text{proj}}\left(u_{\text{IT}}^{\text{new}}\right). \tag{9}$$

To identify related spatial configurations, UEM calculates the similarity between the new key K_{new} and existing keys K_i stored in memory to determine how closely the new experience matches existing memories, as shown in (10). After computing the similarity score S_{sim} for K_{new}, UEM identifies the maximum similarity score, which represents the closest match between K_{new} and stored embeddings, helping UEM decide if the new experience is similar to any existing memories

$$S_{\text{sim}}^{i} = \frac{K_{\text{new}} \cdot K_i}{\|K_{\text{new}}\| \|K_i\|} \tag{10}$$

$$S_{\text{sim}} = \max\left(S_{\text{sim}}^{i}\right) \tag{11}$$

where $S_{\text{sim}}^{i} = [S_{\text{sim}}^{1}, S_{\text{sim}}^{2}, \ldots, S_{\text{sim}}^{n}]$ represents the similarity vector with the ith memory entry and S_{sim} indicate the maximum similarity score.

UEM evaluates the novelty of new experiences using a surprise score S_{sur} (12), which includes global and local variability terms. The global variability measures how different a new experience is from all stored memories by calculating the mean deviation. This term becomes more dominant when a new experience is very dissimilar with the existing ones, signaling novelty. The local variability term captures the distribution characteristics of similarity scores through their standard deviation. When similarity scores vary significantly across different memories, indicated by high standard deviation std in (S_{sim}^{i}), this identifies configurations that partially match existing patterns while introducing novel elements

$$S_{\text{sur}} = \underbrace{\lambda\left(1 - \frac{1}{n}\sum_{i=1}^{n} S_{\text{sim}}^{\text{i}}\right)}_{\text{global variability}} + \underbrace{(1-\lambda)\cdot \text{std}\left(S_{\text{sim}}^{i}\right)}_{\text{local variability}} \tag{12}$$

where λ is a learnable parameters.

If the surprise score S_{sur} indicates high novelty ($S_{\text{sur}} \geq \theta_{\text{sur}}$) and the similarity score S_{sim} is low ($S_{\text{sim}} \leq \theta_{\text{sim}}$), UEM will create a new memory entry for K_{new}, as shown in (13). Here, θ_{sur} is the surprise threshold, and θ_{sim} is the similarity threshold

$$M_{\text{new}} = (K_{\text{new}}, V_{\text{new}}) \tag{13}$$

where $V_{\text{new}} = u_{\text{IT}}^{\text{new}}$ represents the complete multimodal embedding stored as the value component.

If the surprise score indicates low novelty ($S_{\text{sim}} \geq \theta_{\text{sim}}$, $S_{\text{sur}} \leq \theta_{\text{sur}}$), UEM may update an existing memory with information from K_{new}, as it closely resembles a past experience as shown in the following equations:

$$V_i^{\text{new}} = \alpha V_i + (1-\alpha) u_{\text{IT}}^{\text{new}} \tag{14}$$

$$M_i = \left(K_i, V_i^{\text{new}}\right) \tag{15}$$

where α is a weighting factor that determines the influence of the new data on the existing memory.

During the memory update phase, the model computes S_{sim} and S_{sur}. For θ_{sim}, the threshold is typically set to range from 0.5 to 0.9 in increments of 0.1. This range reflects how stringent the model is in considering a new experience similar to an existing memory, with higher values indicating a stricter criterion for memory updates. On the other hand, θ_{sur} is set to range from 0.1 to 0.9, with steps of 0.1, which controls the model's sensitivity to novelty. Higher values of θ_{sur} ensure that only highly surprising or dissimilar experiences will lead to the creation of new memory entries. To identify the optimal configuration, we conduct a grid search that involves systematically evaluating a range of hyperparameter values for θ_{sim} and θ_{sur}.

3) Contextual Scoring: The contextual scoring system builds upon the UEM update mechanism by establishing not only the best match but also how memories relate as an interconnected network. It refines retrieval to acquire a coherent set of related memories by using contextual relevance. This approach helps UEM not only assess individual memories but also make sense of complex, interconnected memory entries in various contexts.

To quantify the contextual relevance between two memory entries M_i and M_j, we compute the contextual score S_{ij}, a measure that captures the similarity between memory keys K_i and K_j (16). The similarity matrix $\bar{S}$ (17) is then calculated for all memory pairs, establishing a complete similarity graph of memory relationships

$$S_{\text{ij}} = \gamma \times \frac{K_i \cdot K_j}{\|K_i\| \|K_j\|} + \delta \tag{16}$$

where γ is a scaling factor that controls the influence of the cosine similarity term and δ is a scalar bias term that adjusts the similarity score

$$\bar{S} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1n} \\ S_{21} & S_{22} & \cdots & S_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ S_{n1} & S_{n2} & \cdots & S_{nn} \end{bmatrix}. \tag{17}$$

Using the similarity matrix $\bar{S}$, we calculate cohesion C_h that ensures internal consistency, preventing fragmentation and outliers and coherence C_r which ensures contextual relevance, enabling memories to interact effectively.

C_h measures the uniformity of pairwise similarity scores across the memory bank, as defined in (18). A low cohesion C_h indicates that memories share consistent similarity relationships, forming a tightly clustered group. This uniformity is critical for tasks requiring precise reasoning, as it ensures that retrieved memories align with the task's context. Coherence C_r measures the average pairwise similarity across all memories, as defined in (19). High coherence reflects strong semantic alignment, where memories share a unified context. This enables effective interaction in few-shot learning, as memories reinforce one another's relevance. By maximizing coherence, the framework ensures that memories collectively emphasize a coherent theme. These two mechanisms ensure that the memory bank is both stable and contextually unified, providing a reliable foundation for tasks requiring precise and coherent

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reasoning

$$C_h = \frac{1}{n(n-1)} \sum_{i \neq j} \left(S_{\text{ij}} - \bar{S}\right)^2 \tag{18}$$

$$C_r = \frac{1}{n(n-1)} \sum_{i \neq j} S_{\text{ij}}. \tag{19}$$

The combined contextual score W_{ij} (20) evaluates memory relationships by integrating three key aspects: immediate semantic relevance through S_{ij}, stability of feature representations through cohesion C_h, and pattern consistency through coherence C_r. This unified scoring helps identify memories that maintain both strong individual matching and stable collective organization, critical for consistent object detection across varying contexts. The score W_{ij} is computed as follows:

$$W_{\text{ij}} = \alpha \cdot S_{\text{ij}} + \beta \cdot C_h + \gamma \cdot C_r \tag{20}$$

where α, β, and γ are the scaling factors that control the relative influence of each component.

This approach establishes the coherent relationships among memories, allowing UEM to organize these contextual scores into a graph structure, where nodes represent individual memory entries and edges denote their contextual relevance. In complex tasks, this framework not only efficiently retrieves the best-matching memory but also acquires contextually related memories, ultimately enhancing the coherence and relevance of the retrieval results. By prioritizing memories that exhibit strong contextual relationships, the model can improve its decision-making capabilities and overall performance.

4) Memory Retrieval: When a query embedding $u_{\text{IT}}^{\text{new}}$ is presented, UEM retrieves relevant entries based on both their similarity to the query and their contextual relevance W_{ij}. The similarity between the current query embedding and each memory key K_i is calculated using the learned similarity function, as shown in (21). Moreover, to prioritize memory entries during retrieval, a relevance weight α_i is computed for each memory M_i as shown in (22)

$$\text{Sim}_R = \sigma\left(\text{Linear}\left(\left[K_{\text{new}} K_i^{\top}\right]\right)\right) \tag{21}$$

$$\alpha_i = \text{SM}\left(\tau \cdot F_i + \text{Sim}_R + \lambda \cdot W_{\text{ij}}\right) \tag{22}$$

where σ indicate the sigmoid, λ is a scaling factor to dynamically adjust W_{ij}, F_i indicates the retrieval frequency score, τ is a scaling factor that controls the influence of F_i, and $[K_i, u_{\text{IT}}]$ represents the concatenation of the memory key and the query embedding.

Without this scaling τ, memories that are frequently accessed could dominate the retrieval process, potentially leading to overfitting. Over reliance on frequently accessed memories would make the model more likely to recall common patterns but less likely to incorporate rarer, contextually important features. By scaling F_i with τ, the model ensures that memories are not only selected based solely on how often they have been accessed but also on their relevance to the current query. This helps the model avoid biasing its retrieval toward frequently seen patterns, thereby promoting diversity in the feature extraction.

The values retrieved from memory, weighted by their corresponding relevance scores α_i, are aggregated to form a contextually enriched memory embedding, as shown in (23). To fully leverage this enriched representation, the memory output m is then combined with the new multimodal input $u_{\text{IT}}^{\text{new}}$ before passing it to the decoder as given in (24)

$$m = \sum_{i=1}^{k} \alpha_i V_i \tag{23}$$

$$m_E = \text{TOPK}\left(\text{MLP}\left(m, u_{\text{IT}}^{\text{new}}\right)\right) \tag{24}$$

where m represents the final embedding that consolidates information from the top k relevant memory entries, while m_E is the final output embedding, integrating the contextually enriched memory with the new multimodal input.

This retrieval approach uses UEM's contextual scoring and similarity measures to refine memory selection and prioritization. The surprise score identifies novel inputs for memory updates, while similarity and relevance weights ensure the retrieval of the most pertinent past experiences, balancing past knowledge with adaptation to new inputs.

C. Memory Augmented Decoder

MAD extends the DETR framework [13] (Fig. 3) by integrating contextual information from CLF and past experiences from UEM to generate memory-augmented queries. DETR generates queries based solely on current input features, which can limit generalization, especially in complex or few-shot detection scenarios. In contrast, MAD utilizes a UEM that stores past experiences in the form of episodic memory, enabling the model to leverage relevant historical context for improved detection.

To create memory-augmented queries q_{memory}, MAD employs a memory retrieval mechanism that generates a contextually enriched memory embedding m_E, as shown in (24). This embedding encapsulates the most relevant historical information, enabling the model to integrate the past knowledge into the current query. These q_{memory} are constructed by applying a projection function f_{query}, which combines the original DETR queries Q_{DETR} with the episodic memory embedding m_E, as shown in (25). This fusion enables the decoder to make predictions informed by both the present visual features and historical context

$$q_{\text{memory}} = f_{\text{query}}(Q_{\text{DETR}}, m_E). \tag{25}$$

These memory-augmented queries incorporate both the current multimodal context and relevant historical information, preparing them for interaction with other components in the model. Each q_{memory} interacts with the encoded image features v_I and episodic memory embeddings m_E through a cross-attention mechanism. The attention weights α_D between q_{memory} and each image feature v_I are calculated to assess the relative importance of each image feature with respect to the query, as shown in the following equation:

$$\alpha_D = \text{SM}\left(\frac{q_{\text{memory}} \cdot v_I}{\sqrt{d}} + b\right) \tag{26}$$

where b is a positional bias that encodes spatial layout information, helping the model localize important features in the image. This attention mechanism allows the model to

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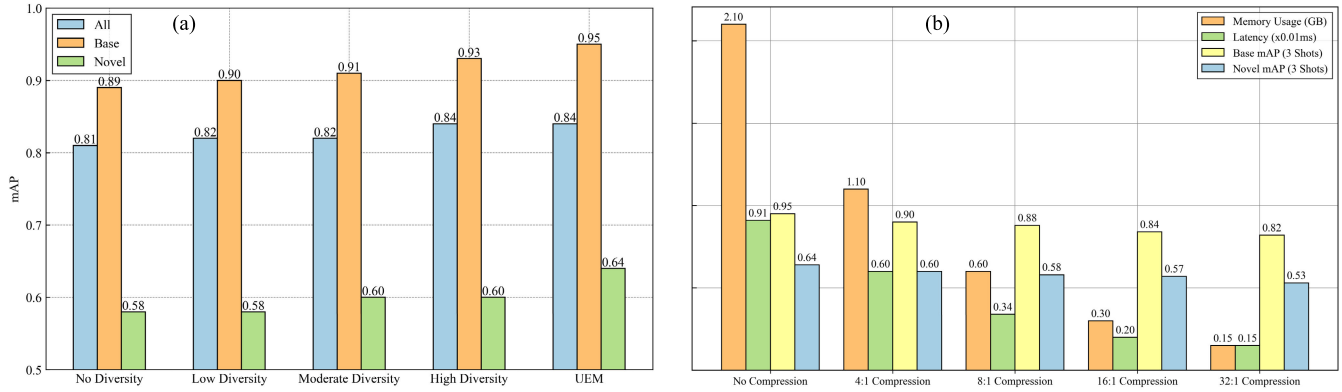


Fig. 6. (a) UEM improves generalization over description-only methods, which use static representations without adaptive memory. (b) Key–value memory compression affects performance, with the uncompressed configuration being best for detection accuracy and generalization, especially for novel class detection.

focus on the most relevant regions of the image when making predictions, thereby improving the accuracy of object detection in complex scenes. With these attention weights, a spatially aligned intermediate query embedding $\hat{q}_{\text{memory}}$ is computed as shown in the following equation:

$$\hat{q}_{\text{memory}} = \sum \alpha_D v_I. \tag{27}$$

These intermediate embeddings $\hat{q}_{\text{memory}}$ are then further refined by attending to contextual embeddings u_{IT} from CLF. To determine the significance of the multimodal embedding in relation to $\hat{q}_{\text{memory}}$, we compute the attention weights β_D as shown in the following equation:

$$\beta_D = \text{SM}\left(\frac{\hat{q}_{\text{memory}} \cdot u_{\text{IT}}}{\sqrt{d}} + b_{\text{ik}}\right) \tag{28}$$

where b_{ik} is a learned contextual bias specific to the memory embeddings, allowing the model to effectively combine visual and textual information for enhanced detection performance. This additional layer of attention facilitates the integration of current contextual features into the query representation, improving the model's contextual awareness and aiding in more accurate object detection.

Finally, the memory-integrated query representation Q_F is calculated as follows:

$$Q_F = \hat{q}_{\text{memory}} + \sum \beta_D m_E. \tag{29}$$

The final memory-augmented queries Q_F are used to obtain the bounding box predictions b_i and class predictions $p_c(i)$, as shown in the following equations, respectively:

$$b_i = \text{MLP}_{\text{bbox}}(Q_F) \tag{30}$$

$$p_c(i) = \text{SM}(\text{MLP}_{\text{class}}(Q_F)). \tag{31}$$

Following DETR's approach, we frame object detection as a direct set prediction task. During training, we utilize a global set-based loss and bipartite matching to precisely align predictions with ground-truth objects.

IV. Experimentation

A. Datasets

1) NWPU VHR-10: NWPU VHR-10 [7] dataset consists of 715 high-resolution RGB images from Google Earth and 85 pan-sharpened color infrared images from the Vaihingen collection. Within these images, 3775 object instances across ten classes are labeled, including airplane, baseball diamond, basketball court, bridge, harbor, ground track field, ship, storage tank, tennis court, and vehicle. In addition, 150 images containing negative samples are included. Table II outlines the novel splits for few-shot learning, while the remaining classes serve as base classes.

2) DIOR: DIOR [62] dataset, a widely used benchmark for object detection in RS, comprises 23 463 images sourced from Google Earth. These images contain 192 472 instances across 20 object classes, including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. The images are standardized to spatial resolutions ranging from 0.5 to 30 m. Table II details the novel splits for few-shot learning, with the remaining classes functioning as base classes.

3) iSAID: iSAID [63] dataset highlights significant domain shifts in aerial imagery, featuring 2806 high-resolution images with 655 451 object instances across 15 classes. It combines RGB imagery from Google Earth and CycloMedia with panchromatic (grayscale) imagery from GF-2 and JL-1 satellites, introducing spectral and textural disparities. The dataset also exhibits resolution variability, geographical diversity (urban, rural, and coastal), and intraclass variance (e.g., regional differences in infrastructure or vehicles), creating substantial contextual and appearance-based domain gaps. These factors, along with fine-grained bounding boxes and instance-level masks, make iSAID a challenging benchmark for evaluating model robustness across diverse imaging conditions and geographies. The base/novel splits (Table II) further isolate distinct subsets, emphasizing cross-domain generalization.

B. Implementation Detail and Evaluation Metrics

MemDeT was implemented using the PyTorch framework and trained on three datasets: NWPU VHR-10, DIOR, and iSAID. The architecture comprises a ViT with $L_V = 8$ layers, BERT with $L_T = 8$ layers, CLF with $L_{\text{CLF}} = 4$ layers, and MAD with $N = 6$ layers, where L denotes the number of

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TABLE II
BASE–NOVEL SPLIT FOR RS DATASETS IN FSOD

Dataset	Split	Novel					Base
DIOR [62]	1	baseball field	basketball court	bridge	chimney	ship	rest
	2	airplane	airport	expressway toll station	harbor	ground track field	rest
	3	dam	golf course	storage tank	tennis court	vehicle	rest
	4	express service area	overpass	stadium	train station	windmill	rest
NWPU-VHR-10 [7]	1	airplane	baseball diamond	tennis court			rest
	2	basketball	ground track field	vehicle			rest
iSAID [63]	1	helicopter	ship	plane	large vehicle		rest
	2	baseball diamond	soccer ball field	roundabout			rest
	3	ground track field basketball court	helicopter	baseball diamond	roundabout	soccer ball field	rest

TABLE III
IMPACT OF MODULES ON PERFORMANCE

MemDeT			3 Shots			5 Shots			10 Shots			Parameters	Flops
MLF	UEM	MAD	Base	Novel	All	Base	Novel	All	Base	Novel	All	(M)	(G)
×	×	×	0.60	0.45	0.68	0.77	0.57	0.72	0.77	0.66	0.74	303	376
✓	×	×	0.82	0.48	0.70	0.82	0.63	0.80	0.82	0.70	0.79	109	158
×	✓	×	0.86	0.53	0.75	0.86	0.68	0.83	0.85	0.73	0.83	231	288
×	✓	✓	0.90	0.57	0.78	0.89	0.72	0.87	0.90	0.78	0.89	260	309
✓	✓	✓	0.95	0.64	0.84	0.95	0.75	0.94	0.96	0.85	0.94	126	239

TABLE IV
MEMDET'S ROBUSTNESS AND GENERALIZATION IN HANDLING CONTEXTUAL DIVERSITY, COMPARED WITH VARIATIONS IN DESCRIPTION ALONE

Diversity	3 Shots			5 Shots			10 Shots		
	Base	Nov	All	Base	Nov	All	Base	Nov	All
None	0.89	0.58	0.81	0.90	0.67	0.82	0.89	0.78	0.88
Low	0.90	0.58	0.82	0.92	0.68	0.83	0.92	0.80	0.88
Mod.	0.91	0.60	0.82	0.93	0.69	0.86	0.92	0.80	0.89
High	0.93	0.60	0.84	0.94	0.71	0.87	0.94	0.81	0.90
UEM	0.95	0.64	0.84	0.95	0.75	0.94	0.96	0.85	0.94

TABLE V
IMPACT OF KEY–VALUE MEMORY COMPRESSION ON PERFORMANCE, REVEALING THE TRADEOFFS BETWEEN STORAGE EFFICIENCY AND MODEL ACCURACY

Comp. Ratio	Mem. (GB)	Lat. (ms)	3 Shots			5 Shots			10 Shots		
			Base	Nov	All	Base	Nov	All	Base	Nov	All
None	2.1	91	0.95	0.64	0.84	0.95	0.75	0.94	0.96	0.85	0.94
4:1	1.1	60	0.90	0.60	0.80	0.90	0.71	0.89	0.91	0.80	0.89
8:1	0.60	34	0.88	0.58	0.77	0.88	0.69	0.85	0.88	0.77	0.87
16:1	0.30	20	0.84	0.57	0.75	0.85	0.66	0.81	0.85	0.73	0.82
32:1	0.15	15	0.82	0.53	0.72	0.81	0.61	0.77	0.83	0.62	0.78

layers for encoders and N denotes the number of layers in the decoder. Input images were resized to 512×512 pixels and normalized. The base model was trained for 100 000 iterations using the Adam optimizer, with an initial learning rate of 0.0005. The learning rate was reduced by a factor of 0.1 at iterations 40 000, 55 000, and 60 000. Few-shot fine-tuning was conducted over 4000 iterations with an initial learning rate of 0.0005, decaying by 0.1 at iterations 1000, 3000, and 4000. The model performance was evaluated using the mean average precision (mAP) metric, focusing on both localization and classification accuracy across the datasets.

C. Ablation Study

In this section, we conduct ablation experiments to assess the contributions of various modules and design choices in our proposed model. We first evaluate the impact of individual modules on overall model performance as shown in Table III. Next, we investigate how variations in textual description affect overfitting and generalization, compared with UEM, detailed in Table IV. To further examine memory efficiency, we assess the impact of key–value memory compression on performance in Table V. In addition, we explore the effectiveness of different memory update mechanisms in Table VI, providing insights into how dynamic updates influence model adaptation. Finally, we conduct an ablation study on memory-augmented queries in MAD, as shown in Table VII, to determine their role in refining feature retrieval and overall detection accuracy. Together, these ablations provide a comprehensive analysis of our model's components and mechanisms, highlighting their contributions to optimizing FSOD.

1) Impact of Modules on Performance: Table III evaluates the impact of each proposed module against the baseline. The baseline model, featuring BridgeTower [61] and DETR decoder [13], achieved an mAP of 0.72. Replacing BridgeTower with CLF boosted the model's mAP to 0.79 by effectively merging multilayer features based on contextual relevance, enriching feature representation. Incorporating UEM further enhanced generalization, achieving mAP scores of 0.75, 0.83, and 0.83 in three-, five-, and ten-shot settings by leveraging past experiences to better handle rare or unseen objects. The integration of CLF and UEM demonstrated even stronger performance, reaching mAPs of 0.78, 0.87, and 0.89 by balancing enhanced feature fusion with improved adaptability to distribution shifts. Finally, MemDeT achieved top performance, with mAPs of 0.84, 0.94, and 0.94 across shot settings. This unified design maintains a manageable parameter count (126 M) and computational load (239 G

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TABLE VI
IMPACT OF MEMORY UPDATE MECHANISM ON PERFORMANCE

Update Mech.	Mem. (GB)	Ret. (ms)	3 Shots			5 Shots			10 Shots		
			Base	Nov	All	Base	Nov	All	Base	Nov	All
Full	3.4	150	0.84	0.54	0.77	0.86	0.65	0.83	0.88	0.80	0.83
Select.	2.4	80	0.60	0.50	0.63	0.60	0.63	0.65	0.61	0.63	0.64
Prior.	2.1	88	0.91	0.60	0.81	0.92	0.70	0.88	0.92	0.81	0.90
Dynamic	2.1	91	0.95	0.64	0.84	0.95	0.75	0.94	0.96	0.85	0.94

TABLE VII
ABLATION ON MEMORY-AUGMENTED QUERIES IN MAD

Config.	3 Shots			5 Shots			10 Shots		
	Base	Nov	All	Base	Nov	All	Base	Nov	All
Disabled	0.77	0.55	0.69	0.78	0.57	0.70	0.74	0.57	0.69
Partial	0.91	0.60	0.79	0.88	0.70	0.85	0.90	0.78	0.88
Enabled	0.95	0.64	0.84	0.95	0.75	0.94	0.96	0.85	0.94

TABLE VIII
ABLATION STUDY OF UEM PARAMETERS ON NWPU VHR-10 DATASET. RESULTS SHOW ACCURACY (%) ON BASE, NOVEL, AND ALL CLASSES ACROSS THREE-, FIVE-, AND TEN-SHOT SETTING. PARAM: PARAMETERS AND VAL: VALUE

Param.	Val.	3-shot			5-shot			10-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All
θ_{sur}	0.3	92.1	61.2	81.4	93.2	72.5	91.8	94.1	82.3	92.4
	0.5	95.0	64.0	84.0	95.0	75.0	94.0	96.0	85.0	94.0
	0.7	93.8	62.5	82.8	94.1	73.2	92.5	95.2	83.1	93.1
	0.9	91.5	60.8	80.2	92.8	71.4	90.6	93.5	81.2	91.8
θ_{sim}	0.3	91.8	60.5	80.8	92.5	71.8	91.2	93.8	82.1	92.0
	0.5	93.2	62.1	82.4	93.8	73.4	92.6	94.5	83.4	92.8
	0.7	95.0	64.0	84.0	95.0	75.0	94.0	96.0	85.0	94.0
	0.9	92.4	61.2	81.5	93.1	72.2	91.5	94.2	82.5	92.2
α	0.2	91.2	60.1	80.4	92.1	71.5	90.8	93.5	81.8	91.6
	0.4	93.5	62.4	82.6	93.8	73.2	92.4	94.8	83.2	92.8
	0.6	95.0	64.0	84.0	95.0	75.0	94.0	96.0	85.0	94.0
	0.8	92.8	61.8	81.8	93.4	72.5	91.8	94.5	82.8	92.4

FLOPs), underscoring the complementary strengths of CLF, UEM, and MAD in optimizing FSOD tasks.

2) Impact of Description Variation on Overfitting: Fig. 6(a) and Table IV illustrate the effectiveness of UEM in enhancing generalization, especially in handling contextual diversity, compared with variations in description alone. To evaluate UEM's impact, we replaced it within MemDeT with four description-only configurations no diversity (none), low diversity (low), moderate diversity (mod), and high diversity (high) to assess the model's adaptability to different levels of contextual variation. The no diversity configuration led to significant base class overfitting, with an mAP of 0.81 in the three-shot setting, but poor novel class generalization (mAP of 0.58). Low diversity, via synonym replacements, slightly improved the mAP to 0.82 but did not prevent overfitting due to limited contextual change. Sentence restructuring, introducing moderate diversity, improved generalization, with an mAP of 0.86 in the five-shot setting. This introduced contextual shifts without altering core content, aiding generalization beyond base classes. High diversity, with varied contexts, yielded the highest description-only mAP (0.90 in the ten-shot setting), enhancing representation adaptability. In comparison, UEM consistently outperformed all description-only methods across all-shot settings, achieving mAP scores of 0.94 and 0.96 in the five- and ten-shot settings, respectively. UEM's dynamic encoding and retrieval of context-rich information provides inherent contextual variation, promoting robust generalization and reducing overfitting. These results highlight UEM's ability to inherently support contextual diversity, facilitating superior generalization across base and novel classes.

3) Impact of Key–Value Memory Compression on Performance: Fig. 6(b) and Table V show the impact of key–value memory compression (Comp. Ratio) on memory usage (Mem), retrieval speed (Lat.), and detection performance, revealing tradeoffs between storage efficiency and model accuracy. Without compression, the model achieves the highest mAP of 0.94 with 2.1 GB of memory usage and a retrieval latency of 91 ms. Applying a 4:1 compression ratio reduces memory usage to 1.1 GB and retrieval latency to 60 ms, though it results in a slight mAP drop to 0.89. Increasing the compression to 8:1 further decreases memory to 0.6 GB and retrieval time to 34 ms, but mAP dips to 0.87, showing incremental tradeoffs in accuracy for storage and speed. At higher compression ratios, such as 16:1 and 32:1, memory usage drops significantly to 0.3 and 0.15 GB, with retrieval times of 20 and 15 ms, respectively. However, these configurations also see substantial declines in mAP (0.82 and 0.78), indicating a loss of critical contextual information. While retrieval speed improves, the marked reduction in accuracy suggests that these higher compression levels may not be optimal for applications, where high detection fidelity is essential. The uncompressed configuration in UEM proves to be the most effective for scenarios prioritizing detection accuracy and generalization, particularly for novel class detection.

4) Impact of Memory Update Mechanism on Performance: Table VI highlights the effect of various memory update mechanisms in UEM on model performance, memory usage (Mem.), and retrieval latency (Lat.). The full replacement approach (Full), which refreshes the entire memory entry at each new task, achieves an mAP of 0.83 with a memory usage of 3.4 GB and a retrieval latency of 150 ms. While this mechanism enhances adaptability to new tasks, its high latency makes it impractical for real-time applications. The selective update mechanism (Select), focusing on novel examples based on surprise alone, reduces memory usage to 2.4 GB and cuts retrieval time to 80 ms but lowers mAP to 0.64, indicating limited generalization to base tasks due to its selective nature. The priority update mechanism (Prior.), which updates memory with contextually relevant examples based on similarity only, achieves a higher mAP of 0.90, using 2.1 GB of memory, and a retrieval time of 88 ms. This approach efficiently prioritizes essential base class information, balancing adaptability with memory efficiency. The dynamic memory update mechanism (dynamic), which combines both surprise and similarity, reaches the highest performance with an mAP of 0.94, using 2.1 GB of memory, and a retrieval latency of 91 ms. This configuration maximizes adaptability and generalization by dynamically updating task-relevant information, offering robust performance across both base and novel classes. Ideal for real-world applications, this mechanism excels in both generalization and rapid adaptation to new data, making it the optimal

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TABLE IX
FSOD PERFORMANCE ON NWPU VHR-10 UNDER (3-, 5-, 10-, AND 20-SHOT SETTINGS) ON TWO DIFFERENT BASE/NOVEL SPLITS

Split	Methods	3 Shots			5 Shots			10 Shots			20 Shots		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	0.83	0.15	0.63	0.83	0.16	0.63	0.84	0.24	0.66	0.83	0.27	0.66
	Meta-RCNN	0.87	0.20	0.67	0.86	0.22	0.67	0.87	0.27	0.69	0.87	0.28	0.70
	FsDetView	0.88	0.25	0.69	0.88	0.30	0.70	0.88	0.32	0.71	0.88	0.33	0.71
	P-CNN	0.83	0.42	0.71	0.83	0.49	0.73	0.83	0.63	0.78	0.84	0.67	0.79
	iMTFA	0.87	0.43	0.74	0.86	0.50	0.75	0.87	0.70	0.82	0.88	0.73	0.83
	Incremental-DETR	0.87	0.51	0.76	0.87	0.53	0.77	0.88	0.70	0.83	0.88	0.73	0.84
	TFA w/cos	0.89	0.09	0.65	0.90	0.09	0.65	0.90	0.09	0.66	0.71	0.11	0.66
	DeRFCN	0.87	0.38	0.73	0.87	0.46	0.75	0.87	0.63	0.80	0.87	0.65	0.80
	TEMO	0.86	0.58	0.78	0.85	0.67	0.80	0.87	0.75	0.83	0.89	0.76	0.85
	G-FSDet	0.89	0.49	0.77	0.88	0.56	0.79	0.88	0.72	0.83	0.90	0.75	0.85
	ST-FSOD	0.90	0.57	0.77	0.90	0.68	0.82	0.91	0.70	0.84	0.91	0.72	0.84
	FSIOD-DFP	0.91	0.55	0.81	0.91	0.67	0.91	0.91	0.75	0.86	0.91	0.77	0.87
	UMFT	0.91	0.58	0.79	0.91	0.68	0.83	0.92	0.79	0.88	0.92	0.79	0.88
	MemDeT	**0.95**	**0.64**	**0.84**	**0.95**	**0.75**	**0.94**	**0.96**	**0.85**	**0.94**	**0.96**	**0.87**	**0.94**
2	FRW	0.85	0.29	0.68	0.86	0.31	0.69	0.85	0.32	0.70	0.87	0.34	0.71
	Meta-RCNN	0.87	0.21	0.67	0.87	0.35	0.72	0.88	0.37	0.72	0.87	0.39	0.73
	FsDetView	0.88	0.39	0.73	0.89	0.40	0.75	0.89	0.45	0.76	0.89	0.46	0.76
	P-CNN	0.81	0.39	0.69	0.81	0.46	0.71	0.81	0.56	0.73	0.81	0.58	0.75
	TFA w/cos	0.90	0.11	0.66	0.91	0.12	0.67	0.91	0.11	0.67	0.90	0.11	0.67
	DeRFCN	0.88	0.39	0.74	0.88	0.46	0.75	0.88	0.54	0.78	0.88	0.57	0.79
	TEMO	0.88	0.51	0.77	0.88	0.59	0.80	0.89	0.65	0.82	0.88	0.73	0.84
	iMTFA	0.87	0.46	0.75	0.88	0.54	0.78	0.88	0.63	0.80	0.88	0.66	0.82
	Incremental-DETR	0.89	0.47	0.77	0.89	0.57	0.80	0.89	0.66	0.82	0.89	0.71	0.84
	G-FSDet	0.89	0.50	0.77	0.90	0.59	0.81	0.89	0.67	0.83	0.91	0.76	0.86
	UMFT	0.91	0.52	0.79	0.90	0.60	0.82	0.90	0.69	0.85	0.91	0.78	0.89
	FSIOD-DFP	0.92	0.52	0.80	0.92	0.61	0.83	0.92	0.71	0.86	0.92	0.78	0.88
	ST-FSOD	0.92	0.54	0.81	0.92	0.62	0.86	0.92	0.73	0.89	0.90	0.79	0.91
	MemDeT	**0.95**	**0.58**	**0.86**	**0.95**	**0.68**	**0.90**	**0.94**	**0.76**	**0.92**	**0.96**	**0.82**	**0.93**

choice for scenarios requiring strong performance and quick responsiveness.

5) Ablation on Memory-Augmented Queries in MAD: Table VII illustrates the effect of memory-augmented queries on MAD's performance. Disabling memory-augmented queries (Disabled) leads to a substantial drop, with the base class mAP declining to 0.77 and the novel class mAP falling to 0.55 in the three-shot setting, demonstrating poor generalization across both base and novel classes. When only partial dependency (Partial) on memory-augmented queries is applied, the model shows a marked improvement, achieving an mAP of 0.91 for the base class and 0.60 for the novel class in the three-shot setting, showing enhanced performance due to limited memory integration. With full memory-augmented queries enabled (Enabled), the model reaches its highest accuracy, achieving an mAP of 0.95 for the base class and 0.64 for the novel class in the three-shot configuration, and maintaining strong results in the five-shot and ten-shot settings, with mAPs of 0.94 (base) and 0.85 (novel) in the ten-shot setting. These findings confirm that comprehensive memory utilization provides essential context, significantly improving object recognition and generalization across novel instances.

6) Impact of UEM Parameters on Performance: Table VIII demonstrates the impact of UEM's key parameters on model performance. The surprise threshold θ_{sur} controls novel class adaptation. At optimal value ($\theta_{\text{sur}} = 0.5$), the model maintains strong base performance (95.0%–96.0%) while achieving high-novel class accuracy (64.0%–85.0% across 3–10 shots). Lower values cause base class degradation (2.9% drop at $\theta_{\text{sur}} = 0.3$), while higher values limit novel class learning (3.8% drop at $\theta_{\text{sur}} = 0.9$). The similarity threshold $\theta_{\text{sim}} = 0.7$ enables effective knowledge transfer, shown by steady novel class improvement (64.0%–85.0%) with increasing shots. In low-shot setting memory update weight $\alpha = 0.6$, balances base knowledge preservation with novel class adaptation. This optimal configuration ($\theta_{\text{sur}} = 0.5$, $\theta_{\text{sim}} = 0.7$, and $\alpha = 0.6$) maintains consistent base performance while enabling significant novel class improvements, demonstrating UEM's ability to prevent forgetting while supporting few-shot learning.

D. Comparison With State of the Art

The proposed MemDeT is compared with several state-of-the-art FSOD methods, including FRW [17], Meta-RCNN [15], FsDetView [64], P-CNN [19], [39], DEFRCN [65], TEMO [21], G-FSDet [24], iMTFA [66], Incremental-DETR [26], FSIOD-DFP [23] and UMFT [20], BOSS [29], ST-FSOD [27], TH-FSOD [28], and BridgeTower [61] on NWPU VHR-10, DIOR, and iSAID datasets.

1) Experiments on NWPU VHR-10: Fig. 7 and Table IX demonstrate MemDeT's effectiveness in handling varying levels of contextual complexity and sample distributions in NWPU VHR-10. Novel classes in Split 1 present distinct challenges: "airplane" exhibits high contextual complexity through systematic arrangements (aircraft–runway alignments and terminal relationships), while sports facilities like "baseball diamond" and "tennis court" show moderate complexity with fixed layouts but varying orientations. These classes

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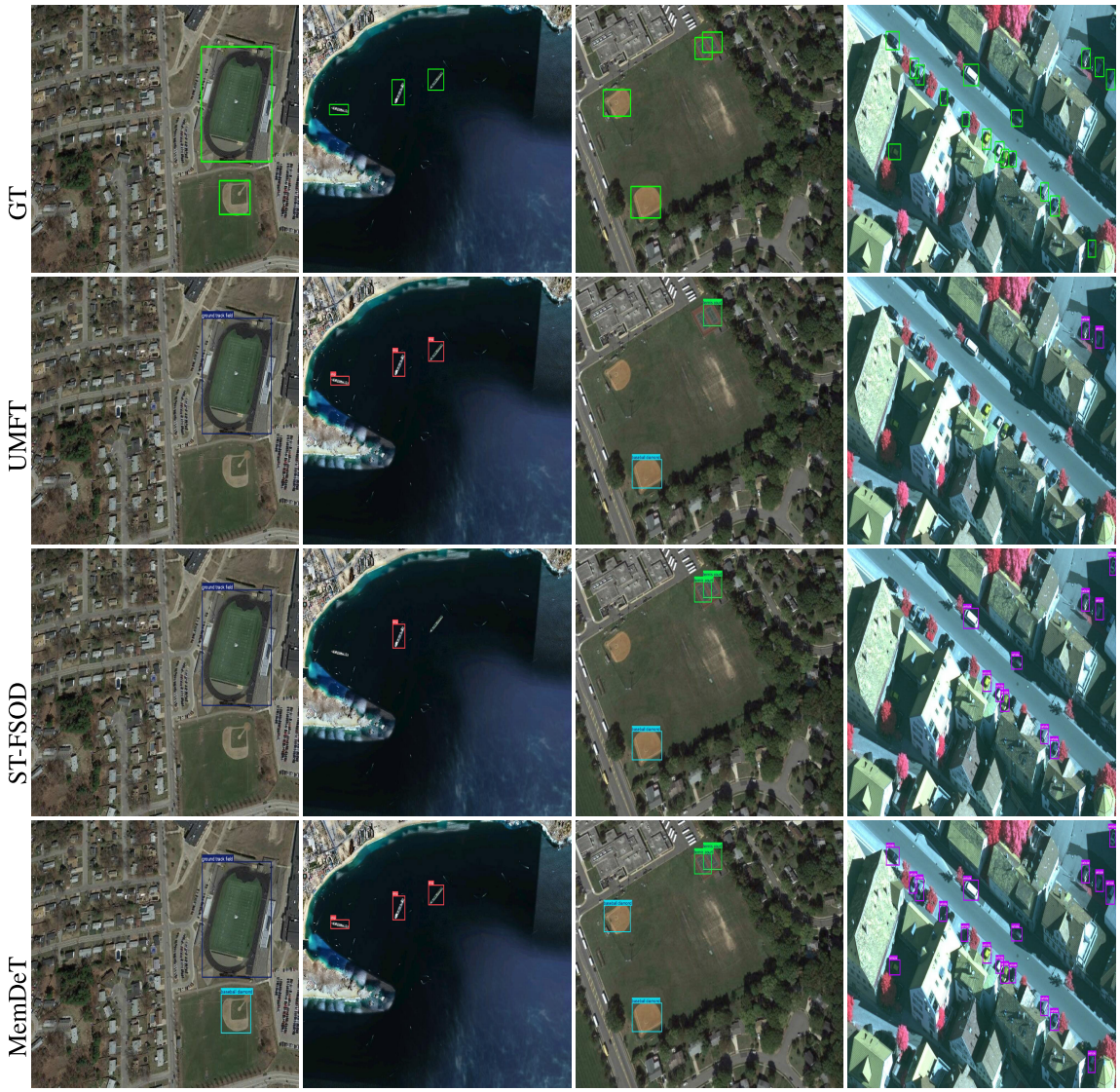


Fig. 7. FSOD visualization on NWPU VHR-10 dataset under ten-shot setting.

also present contrasting distribution patterns, from densely clustered aircraft to isolated sports facilities. CLF processes the image–text features hierarchically, capturing both individual object characteristics and complex spatial relationships. UEM adapts its memory organization to handle varying distribution patterns, while maintaining consistent feature representation across contexts. Whereas, models like UMFT and TEMO rely solely on image–text mechanisms for novel-class detection (0.58) but struggle with knowledge retention, limiting their base class performance (0.91 and 0.86, respectively). Conversely, FSIOD-DFP, an incremental learning-based method focusing on representation stability, effectively retains prior knowledge and maintains stable base performance (0.91), though it lacks the contextual depth from text-based information, resulting in lower novel class accuracy (0.55). In contrast, MemDeT bridges this gap, achieving superior results in both base (0.95) and novel (0.64) classes, demonstrating a strong balance between stability and adaptation. This advantage is particularly pronounced in the 20-shot setting, where MemDeT reaches 0.96 for base and 0.87 for novel classes.

2) Experiments on DIOR: Fig. 8 and Table X demonstrate the MemDeT's capability in handling unique contextual relationships and sample distribution patterns present in DIOR. For instance, in Split 1, bridges and chimneys do not exist in isolation but form part of interconnected networks: bridges connect transportation systems (roads and railways) while chimneys are elements of industrial complexes. This creates complex contextual chains where accurate detection requires understanding both immediate surroundings and broader system relationships. Moreover, it also presents challenging sample distribution characteristics. Industrial structures like chimneys exhibit highly variable density patterns, they can appear as densely packed clusters in industrial zones or as isolated structures in smaller facilities. Sports facilities show structured but incomplete patterns, where supporting infrastructure (parking lots and access roads) may be partially visible or occluded, breaking typical contextual cues. MemDeT's architecture addresses these nested dependencies and variable distributions through a unified fusion and knowledge retrieval mechanism. CLF module processes

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TABLE X
FSOD PERFORMANCE ON DIOR DATASET UNDER (3-, 5-, 10-, AND 20-SHOT SETTINGS) ON FOUR DIFFERENT BASE/NOVEL SPLITS

Split		3 Shots			5 Shots			10 Shots			20 Shots		
	Methods	Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	Meta-RCNN	0.61	0.12	0.48	0.62	0.13	0.50	0.62	0.14	0.50	0.63	0.14	0.51
	TFA w/cos	0.70	0.11	0.55	0.70	0.11	0.56	0.70	0.15	0.57	0.71	0.18	0.58
	DeRFCN	0.62	0.28	0.53	0.62	0.30	0.54	0.61	0.32	0.54	0.62	0.35	0.55
	TEMO	0.61	0.23	0.52	0.64	0.31	0.56	0.65	0.35	0.58	0.65	0.40	0.59
	iMTFA	0.60	0.23	0.51	0.63	0.25	0.53	0.66	0.29	0.57	0.62	0.35	0.56
	Incremental-DETR	0.66	0.27	0.56	0.66	0.30	0.57	0.70	0.36	0.62	0.70	0.38	0.62
	G-FSDet	0.69	0.27	0.58	0.69	0.30	0.59	0.69	0.37	0.61	0.67	0.40	0.62
	FSIOD-DFP	0.72	0.29	0.61	0.72	0.31	0.62	0.72	0.38	0.63	0.72	0.41	0.64
	UMFT	0.72	0.30	0.61	0.73	0.32	0.63	0.73	0.38	0.64	0.72	0.40	0.63
	BOSS	0.72	0.20	0.59	0.75	0.28	0.64	0.77	0.40	0.68	0.79	0.45	0.71
	ST-FSOD	0.73	**0.42**	0.66	0.78	0.46	0.66	0.73	**0.50**	0.67	0.73	0.54	0.68
	TH-FSOD	**0.78**	0.38	0.68	0.79	0.44	0.70	0.80	0.46	0.71	0.79	0.49	0.72
	MemDeT	**0.78**	**0.42**	**0.72**	**0.80**	**0.50**	**0.72**	**0.82**	**0.50**	**0.73**	**0.81**	**0.55**	**0.78**
2	Meta-RCNN	0.62	0.09	0.49	0.63	0.11	0.50	0.63	0.15	0.51	0.64	0.17	0.52
	TFA w/cos	0.71	0.06	0.54	0.71	0.08	0.55	0.70	0.09	0.54	0.70	0.12	0.55
	DeRFCN	0.61	0.14	0.49	0.62	0.16	0.50	0.61	0.18	0.51	0.63	0.21	0.52
	TEMO	0.61	0.16	0.50	0.65	0.18	0.53	0.63	0.21	0.52	0.61	0.23	0.52
	iMTFA	0.6	0.13	0.48	0.60	0.15	0.49	0.60	0.19	0.50	0.61	0.23	0.52
	Incremental-DETR	0.67	0.13	0.54	0.67	0.15	0.54	0.67	0.20	0.55	0.68	0.20	0.56
	G-FSDet	0.69	0.14	0.55	0.69	0.16	0.57	0.68	0.20	0.56	0.68	0.22	0.57
	FSIOD-DFP	0.70	0.18	0.57	0.70	0.20	0.57	0.70	0.22	0.58	0.70	0.24	0.58
	UMFT	0.70	0.19	0.58	0.71	0.23	0.59	0.71	0.22	0.60	0.72	0.26	0.61
	BOSS	0.70	0.09	0.55	0.74	0.16	0.60	0.78	0.32	0.67	0.80	**0.43**	0.71
	ST-FSOD	0.73	0.18	0.59	0.73	0.21	0.60	0.72	0.27	0.61	0.73	0.33	0.63
	TH-FSOD	0.77	0.17	0.62	0.78	0.20	0.64	0.79	0.29	0.67	0.78	0.35	0.67
	MemDeT	**0.79**	**0.26**	**0.67**	**0.81**	**0.25**	**0.69**	**0.81**	**0.33**	**0.70**	**0.82**	0.38	**0.72**
3	Meta-RCNN	0.62	0.10	0.49	0.63	0.12	0.50	0.62	0.12	0.50	0.65	0.16	0.53
	TFA w/cos	0.72	0.08	0.56	0.71	0.10	0.56	0.72	0.10	0.57	0.73	0.18	0.59
	DeRFCN	0.63	0.15	0.51	0.63	0.18	0.52	0.64	0.20	0.53	0.64	0.25	0.54
	TEMO	0.66	0.16	0.53	0.66	0.23	0.55	0.68	0.25	0.57	0.68	0.31	0.58
	iMTFA	0.64	0.16	0.52	0.64	0.20	0.53	0.66	0.23	0.55	0.66	0.28	0.56
	Incremental-DETR	0.68	0.16	0.55	0.69	0.23	0.57	0.69	0.25	0.58	0.68	0.31	0.59
	G-FSDet	0.71	0.16	0.57	0.70	0.23	0.58	0.71	0.26	0.60	0.71	0.32	0.61
	FSIOD-DFP	0.73	0.17	0.59	0.73	0.25	0.61	0.73	0.30	0.62	0.73	0.32	0.62
	UMFT	0.72	0.20	0.60	0.74	0.26	0.62	0.74	0.30	0.63	0.74	0.34	0.64
	BOSS	0.71	0.10	0.56	0.77	0.14	0.61	0.78	0.29	0.66	**0.80**	0.35	0.69
	ST-FSOD	0.75	0.21	0.62	0.76	0.26	0.63	0.76	0.31	0.65	0.76	0.35	0.65
	TH-FSOD	**0.76**	0.25	0.63	0.76	0.26	0.64	0.77	0.32	0.66	0.77	0.38	0.67
	MemDeT	**0.76**	**0.26**	**0.67**	**0.79**	**0.30**	**0.68**	**0.80**	**0.34**	**0.70**	**0.80**	**0.40**	**0.70**
4	Meta-RCNN	0.62	0.14	0.50	0.6	0.16	0.51	0.62	0.15	0.50	0.63	0.18	0.52
	TFA w/cos	0.69	0.10	0.54	0.69	0.14	0.55	0.68	0.14	0.55	0.69	0.12	0.54
	DeRFCN	0.60	0.11	0.47	0.60	0.18	0.49	0.59	0.21	0.50	0.60	0.27	0.52
	TEMO	0.66	0.17	0.54	0.66	0.21	0.55	0.67	0.27	0.57	0.67	0.32	0.58
	iMTFA	0.63	0.15	0.51	0.60	0.20	0.50	0.65	0.24	0.55	0.62	0.31	0.54
	Incremental-DETR	0.67	0.14	0.54	0.66	0.21	0.55	0.64	0.26	0.55	0.61	0.28	0.57
	G-FSDet	0.69	0.17	0.56	0.68	0.21	0.56	0.68	0.26	0.58	0.67	0.32	0.58
	FSIOD-DFP	0.70	0.22	0.57	0.71	0.24	0.58	0.72	0.30	0.60	0.72	0.34	0.61
	UMFT	0.71	0.22	0.58	0.72	0.26	0.60	0.73	0.31	0.62	0.73	0.35	0.63
	BOSS	0.68	0.12	0.54	0.74	0.26	0.62	0.79	0.31	0.67	**0.81**	0.40	**0.70**
	ST-FSOD	0.73	0.20	0.60	0.74	0.25	0.61	0.74	0.33	0.64	0.74	0.38	0.65
	TH-FSOD	0.76	0.16	0.61	0.77	0.27	0.65	0.79	0.30	0.66	0.78	0.40	0.69
	MemDeT	**0.78**	**0.23**	**0.65**	**0.80**	**0.29**	**0.67**	**0.81**	**0.36**	**0.69**	**0.81**	**0.42**	**0.70**

contextual information hierarchically, capturing both local object features and their roles in larger systems (bridge-road connections and chimney-facility relationships). UEM enhances this by maintaining separate memory representations for different contextual scenarios, allowing objects to be recognized whether they appear in dense industrial clusters or as isolated structures. MAD then uses this contextual knowledge to generate adaptive queries that account for both immediate surroundings and broader system relationships. This approach achieves 0.78/0.42 mAP (base/novel) by effectively handling both challenges. Contextual learning methods like UMFT (0.72/0.40) rely solely on direct feature-text alignment, which struggles with nested dependencies, while they can learn immediate object relationships, they fail to maintain stable representations of how these relationships vary across different functional systems. Whereas incremental learning approaches like FSIOD-DFP (0.72/0.29), while good for stability, prevent adaptation to varying instance novel distributions.

3) Experiments on iSAID: Fig. 9 and Table XI show how MemDeT effectively handles the significant sample distribution variations in iSAID. These variations arise from the dataset combining RGB imagery from Google Earth and CycloMedia with panchromatic imagery from GF-2 and JL-1 satellites, creating fundamental shifts in how objects appear and their feature characteristics. MemDeT integrates visual

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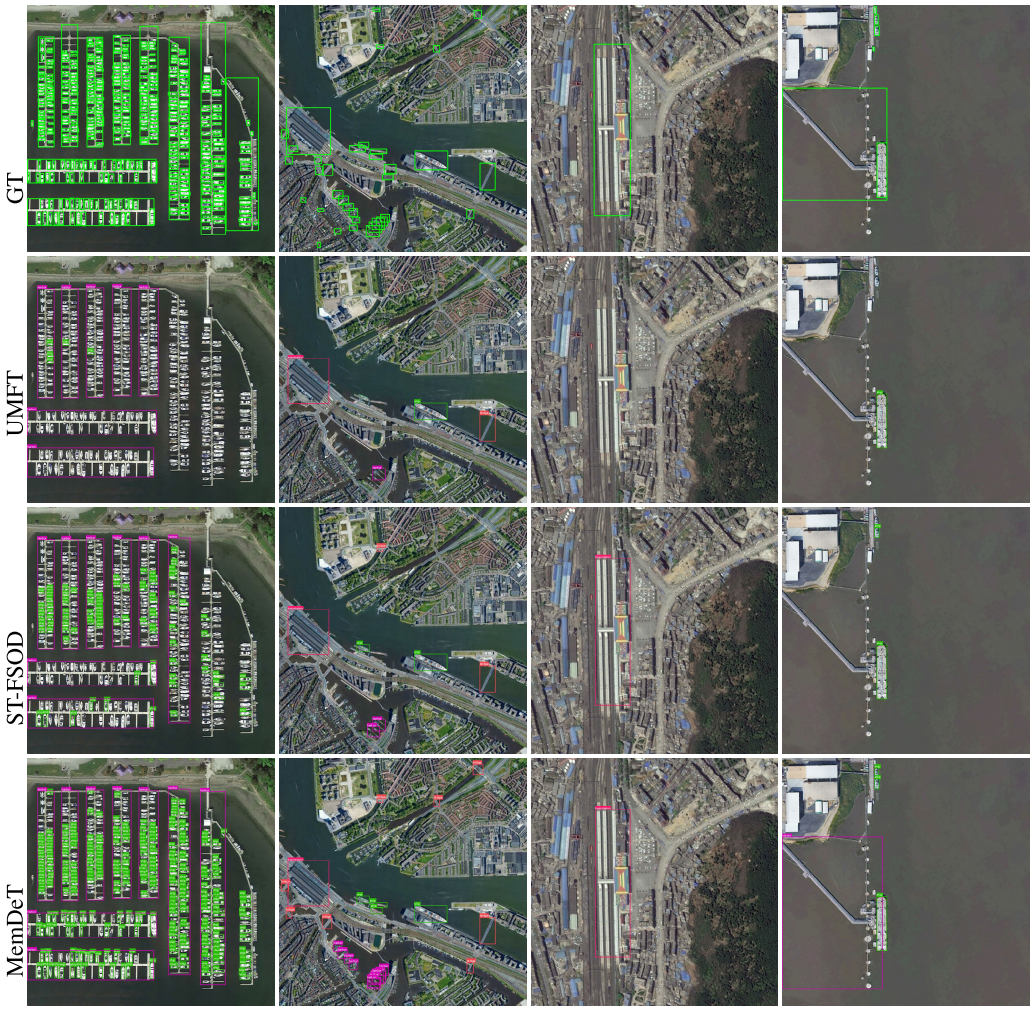


Fig. 8. FSOD visualization on DIOR dataset under ten-shot setting.

and textual information to address these domain-induced distribution shifts. CLF maintains robust feature representation by progressively fusing visual and textual information, CLF maintains robust feature representation by progressively fusing visual and textual information. For visual features, it processes domain-specific characteristics (spectral in RGB and textural in panchromatic), while simultaneously extracting semantic information from text descriptions. This allows CLF to compensate for weakened visual features in panchromatic images with stronger semantic understanding, and vice versa, maintaining detection reliability across sensor types. UEM enhances adaptation by building unified representations that capture both visual and semantic patterns. When encountering sensor variations, UEM's memory entries preserve critical visual features (shape and structure) while maintaining semantic consistency through textual information. This balanced representation ensures that the model can effectively recognize objects even when either visual or textual features become less reliable. In contrast, unimodal methods like ST-FSOD struggle with sensor variations due to their reliance on visual features alone, while UMFT's direct feature-text alignment cannot effectively handle varying feature reliability. MemDeT's superior performance demonstrates the effectiveness of its balanced multimodal approach.

E. Analysis of Model Stability and Cross-Split Performance in FSOD

FSOD in RS presents a significant challenge in balancing feature learning with representation stability. Figs. 10 and 11 illustrate our comprehensive analysis of the NWPU VHR-10 and DIOR datasets, highlighting distinct strategies employed by different methods to navigate this challenge. The contextual learning method UMFT, showcases robust feature learning capabilities, achieving notable improvements in novel class performance (NWPU VHR-10: 0.58–0.79 and DIOR: 0.19–0.26) due to its rich contextual understanding. However, the substantial gap in base class performance reveals difficulties in maintaining stable feature representations. In contrast, the incremental learning approach FSIOD, prioritizes stability, ensuring steady base class performance while exhibiting more conservative gains in novel class detection (NWPU VHR-10: 0.55–0.77 and DIOR: 0.18–0.24), suggesting limitations in its ability to adapt features effectively. TH-FSOD attempts to bridge the gap between these approaches but experiences catastrophic forgetting, evidenced by a slight decline in base performance on DIOR (0.77–0.78) despite improvements in novel classes (0.17–0.35). This indicates instability in feature representations during

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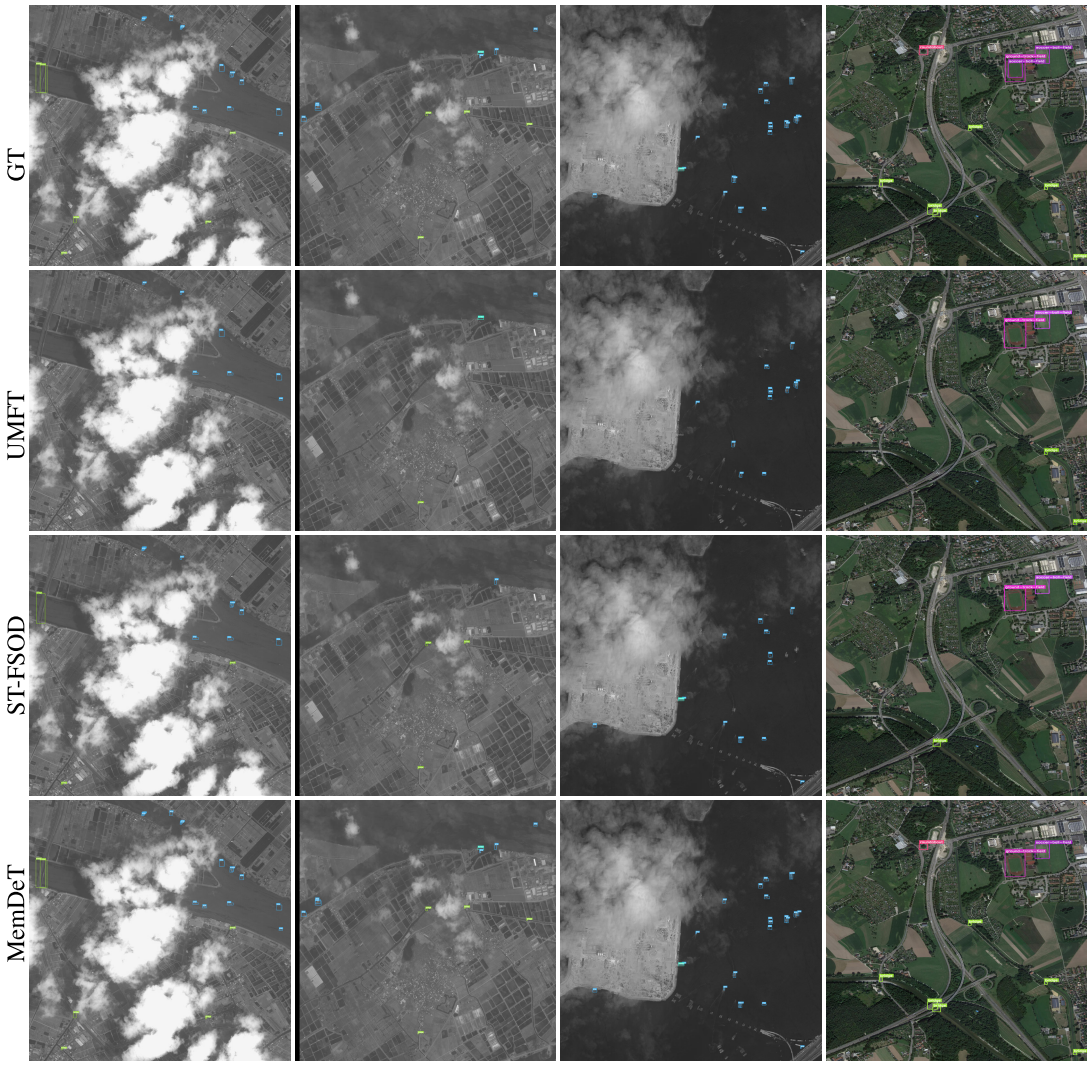


Fig. 9. FSOD visualization on iSAID dataset under 100-shot setting.

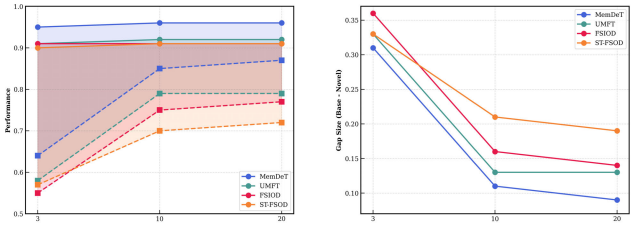


Fig. 10. Performance comparison of FSOD methods on NWPU VHR-10 dataset. (Left) Base (solid lines) and novel (dashed lines) class performance showcasing MemDeT's balanced improvement. (Right) Base–novel gap analysis demonstrating effective knowledge transfer while maintaining stability.

adaptation. MemDeT effectively overcomes these challenges with its memory-augmented architecture, achieving superior adaptation for novel classes while maintaining strong base performance across both datasets (NWPU VHR-10 base: 0.95–0.96 and novel: 0.64–0.87; and DIOR base: 0.79–0.82 and novel: 0.26–0.38). These results show that while contextual and incremental learning approaches excel in their specific strengths, feature adaptation and stability, MemDeT's unified approach successfully addresses both aspects, demonstrating its effectiveness in FSOD.

Fig. 12 provides a comprehensive assessment of model performance across various learning paradigms. The first row compares MemDeT with the image–text method (TEMO), showing that while both excel in novel class detection, MemDeT shows superior coverage across all metrics, particularly in base class performance. The second row compares MemDeT with incremental learning methods, including iMTFA, Incremental-DETR, G-FSDet, and FSIOD-DFP. While these methods are specifically designed to prioritize stability, they still show less effectiveness in overall performance compared with MemDeT's balanced feature learning approach. The third row compares MemDeT with leading state-of-the-art methods, such as TH-FSOD, ST-FSOD, and BOSS, where MemDeT's larger coverage area emphasizes the effectiveness of its unified memory-augmented approach in balancing performance between base and novel classes. Notably, across all three comparisons, it maintains the largest and most symmetric coverage area, indicating consistent performance improvements without sacrificing stability for base classes or adaptability for novel classes.

The extended shot analysis presented in Fig. 13 (3 → 50 shots) offers deeper insights into the model's long-term behavior and stability, demonstrating that MemDeT consis-

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TABLE XI

FSOD PERFORMANCE ON iSAID UNDER 10-, 50-, AND 100-SHOT SETTINGS ACROSS THREE BASE/NOVEL SPLITS

Split	Method	10-shot			50-shot			100-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All
1	TFA	0.59	0.03	0.48	0.48	0.05	0.50	0.61	0.05	0.50
	BridgeTower	0.60	0.07	0.52	0.61	0.15	0.53	0.58	0.15	0.50
	DH-FSDet	0.65	0.05	0.52	0.65	0.13	0.53	0.65	0.17	0.54
	UMFT	0.63	0.08	0.53	0.62	0.20	0.54	0.62	0.32	0.55
	ST-FSOD	0.64	0.10	0.54	0.63	0.25	0.56	0.63	0.34	0.57
	MemDeT	**0.70**	**0.20**	**0.62**	**0.73**	**0.33**	**0.66**	**0.74**	**0.41**	**0.69**
2	TFA	0.57	0.03	0.44	0.59	0.12	0.49	0.59	0.14	0.50
	BridgeTower	0.63	0.06	0.52	0.63	0.15	0.54	0.61	0.17	0.54
	DH-FSDet	0.65	0.15	0.55	0.65	0.29	0.58	0.65	0.36	0.59
	UMFT	0.63	0.16	0.54	0.62	0.38	0.56	0.63	0.45	0.57
	ST-FSOD	0.63	0.18	0.54	0.63	0.39	0.58	0.63	0.45	0.59
	MemDeT	**0.69**	**0.25**	**0.60**	**0.73**	**0.46**	**0.62**	**0.73**	**0.50**	**0.66**
3	TFA	0.59	0.04	0.48	0.61	0.06	0.50	0.62	0.05	0.51
	BridgeTower	0.57	0.09	0.50	0.58	0.23	0.56	0.57	0.26	0.57
	DH-FSDet	0.66	0.10	0.54	0.67	0.20	0.58	0.68	0.23	0.59
	UMFT	0.65	0.12	0.55	0.65	0.30	0.58	0.64	0.34	0.61
	ST-FSOD	0.66	0.14	0.56	0.66	0.31	0.59	0.66	0.33	0.60
	MemDeT	**0.72**	**0.22**	**0.63**	**0.74**	**0.39**	**0.66**	**0.77**	**0.40**	**0.67**

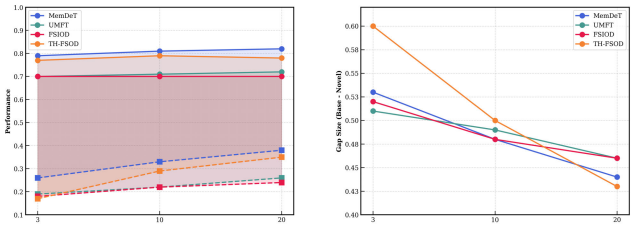


Fig. 11. Performance comparison of FSOD methods on the DIOR dataset. (Left) Base (solid lines) and novel (dashed lines) class performance, showcasing MemDeT's consistent improvement. (Right) Base–novel gap analysis demonstrating model stability.

TABLE XII

CROSS-DATASET EVALUATION RESULTS COMPARING MEMDET WITH EXISTING FSOD METHODS. THE MAP RESULTS FOR COMPETING METHODS ARE SOURCED FROM [27]. ALL METHODS ARE EVALUATED ON THE iSAID NOVEL CLASSES, BUT THEY DIFFER IN THEIR PRETRAINING DATASETS

Shots	Methods	Pretraining	Novel Classes		
			split1	split2	split3
10	FSDetView [65]	iSAID	0.01	0.09	0.05
	TFA [19]	iSAID	0.03	0.09	0.04
	DH-FSDet [68]	iSAID	0.05	0.15	0.10
	ST-FSOD [27]	iSAID	0.10	0.18	0.14
	MemDeT	DIOR	0.13	0.21	0.17
50	FSDetView [65]	iSAID	0.07	0.27	0.17
	TFA [19]	iSAID	0.05	0.12	0.06
	DH-FSDet [68]	iSAID	0.13	0.29	0.20
	ST-FSOD [27]	iSAID	0.25	0.39	0.31
	MemDeT	DIOR	0.28	0.41	0.34
100	FSDetView [65]	iSAID	0.10	0.33	0.24
	TFA [19]	iSAID	0.05	0.14	0.05
	DH-FSDet [68]	iSAID	0.17	0.36	0.23
	ST-FSOD [27]	iSAID	0.34	0.45	0.33
	MemDeT	DIOR	0.35	0.46	0.38

tently improves in both base and novel class performance. The gradually narrowing confidence bands (indicated by blue shading) reflect increasing prediction stability as more shots become available. This stable improvement trend, without performance plateaus or declines, validates the robustness of MemDeT's memory-augmented architecture in accumulating and preserving knowledge. The smooth upward trajectory in novel class performance (red plot), combined with maintained base class stability (blue plot), indicates effective feature adaptation without compromising existing representations, even as data availability increases. This extended analysis underscores MemDeT's ability to leverage additional examples while ensuring stable performance, a critical characteristic for practical RS applications where data availability may vary.

While contextual learning methods excel in feature adaptation and incremental learning approaches focus on stability, MemDeT effectively integrates both through its memory-augmented architecture, demonstrating consistent performance improvements across various shot settings and datasets. Extensive evaluations, including performance gap analysis, radar plots, and extended shot experiments, confirm MemDeT's ability to maintain stable base class performance while adapting to novel classes, making it a robust solution for FSOD in RSI.

F. Cross-Dataset Generalization

To assess MemDeT's capability to generalize across various scenarios, we conduct cross-dataset experiments centered on knowledge transfer to the iSAID [63] dataset. While baseline methods use iSAID for base training, we instead train MemDeT on DIOR base classes and evaluate its performance on novel classes from iSAID to test true cross-dataset generalization. Generalization from DIOR to iSAID presents unique challenges beyond typical domain adaptation. While DIOR provides RGB imagery with consistent resolution and viewing conditions, iSAID introduces dramatic shifts through its multisensor imagery combining RGB and panchromatic data. First, the fundamental feature representation changes, objects that exhibit clear spectral and textural patterns in DIOR appear with significantly reduced information in iSAID's panchromatic images. Common features learned from DIOR's RGB domain, such as color-based object boundaries or spectral signatures, become unreliable or absent in iSAID's panchromatic views. Second, the contextual relationships learned from DIOR may not transfer effectively to iSAID due to different imaging characteristics. Objects that appear clearly separated in DIOR's RGB imagery may blend together in iSAID's panchromatic views, challenging learned spatial dependencies. MemDeT overcomes these cross-dataset challenges through its integrated architecture. CLF's layerwise fusion handles the feature representation shifts by progressively building robust features across transformer layers. In lower layers, it focuses on domain-invariant characteristics such as shapes and structural patterns that remain consistent between RGB and panchromatic imagery. Higher layers then integrate semantic information from text descriptions, which provide stable anchor points regardless of visual domain changes. This hierarchical approach helps bridge the gap between DIOR's rich RGB features and iSAID's reduced panchromatic information. UEM's surprise-driven memory mechanism adapts to the new domain characteristics without forgetting DIOR-learned patterns. When encountering iSAID's panchromatic images, it identifies novel patterns through the surprise score while maintaining relevant RGB-learned features through the similarity score. This balanced approach allows UEM to build a comprehensive feature representation that works across

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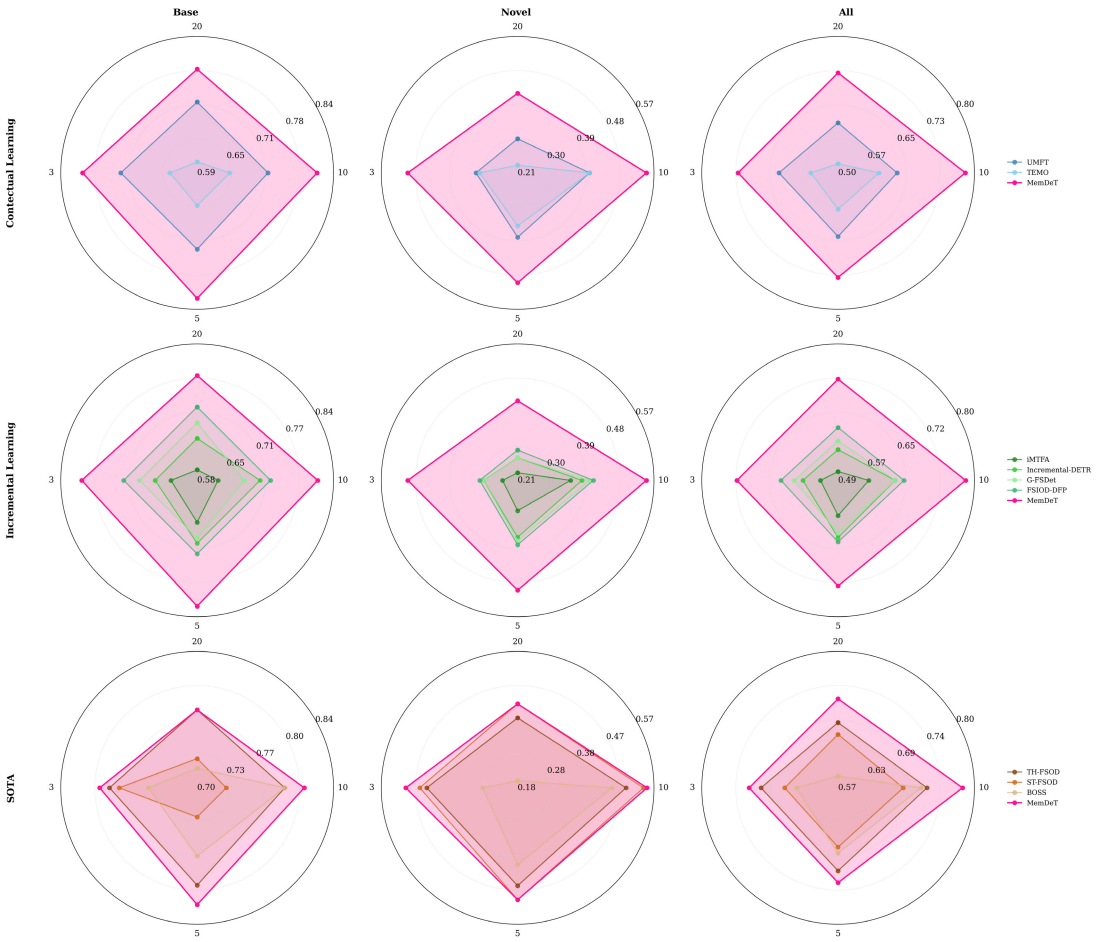


Fig. 12. Comprehensive model comparison through radar plots across different learning paradigms. The top row compares image–text methods, the middle row analyzes incremental learning approaches, and the bottom row evaluates state-of-the-art methods. The columns display base, novel, and overall performance metrics, respectively.

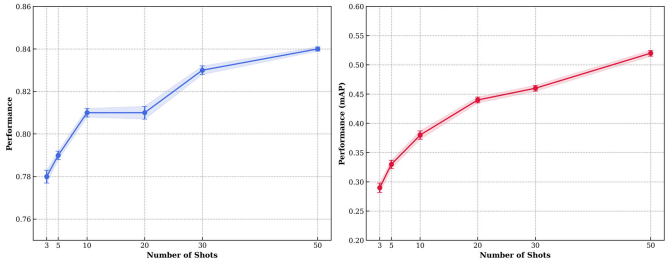


Fig. 13. Extended shot analysis of MemDeT's performance from 3 to 50 shots. (Left) Base class performance trajectory with confidence bands showing stability improvement. (Right) Novel class adaptation demonstrating consistent learning capability. The narrowing confidence bands reflect increasing prediction stability as more shots are available.

both domains. Rather than treating panchromatic features as entirely new, it links them with related RGB patterns stored during DIOR training. MAD enhances cross-dataset detection by generating adaptive queries that work in both domains. It combines preserved RGB-domain knowledge from DIOR with newly learned panchromatic patterns from iSAID, enabling robust object detection despite the significant domain gap. The memory-augmented queries help maintaining spatial relationship understanding even when visual characteristics change dramatically between datasets. Table XII illustrates that MemDeT effectively transfers learned contextual relationships while maintaining stable performance, even in this more challenging setup. The significant performance gap between MemDeT and baseline methods, especially in low-shot scenarios ten shots: 0.13–0.21 mAP versus FSDetView: mAP of 0.01–0.09), demonstrates that our memory-augmented architecture successfully preserves and adapts learned contextual patterns across various RS domains. This advantage continues in higher shot settings, with MemDeT showing consistent improvements at 50 shots (mAP of 0.28–0.41) and 100 shots (mAP of 0.35–0.46). These results validate our hypothesis that a memory-augmented approach can effectively bridge the gap between maintaining stable feature representations and adapting to new contextual relationships, even when transferring knowledge between distinct RS scenarios.

V. Conclusion

While existing approaches face the fundamental challenge of balancing contextual feature learning with representation stability. Contextual learning models struggle with catastrophic forgetting, whereas incremental learning models prioritize maintaining stable feature representations at the cost of contextual understanding. MemDeT addresses these challenges by integrating three modules that balance object features and context through CLF, prevent catastrophic forgetting with UEM, and ensure robust detection with limited examples

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using MAD. Evaluations on NWPU VHR-10, DIOR, and iSAID show that MemDeT outperforms existing methods, maintaining performance on base classes and adapting to novel classes. Moreover, cross-dataset experiments demonstrate its strong generalization capabilities, advancing FSOD for RS. Future work can focus on optimizing memory mechanisms for more efficient knowledge retention and exploring cross-domain adaptation.

REFERENCES

[1] H. Wang, X. Wang, and Y. Cheng, "Graph meta transfer network for heterogeneous few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501112.

[2] W. Wang, Y. Sun, J. Li, and X. Wang, "Frequency and spatial based multi-layer context network (FSCNet) for remote sensing scene classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103781.

[3] L. Zhang, Z. Tan, G. Zhang, W. Zhang, and Z. Li, "Learn more and learn usefully: Truncation compensation network for semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4403814.

[4] A. Siddique, Z. Li, A. Azeem, Y. Zhang, and B. Xu, "Multiscale context-aware feature fusion network for land-cover classification of urban scene imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8475–8491, 2023.

[5] H. Wang, S. Qu, Z. Qiao, and X. Liu, "KCDNet: Multimodal object detection in modal information imbalance scenes," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.

[6] X. Yang, S. Zhang, S. Duan, and W. Yang, "An effective and lightweight hybrid network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600711.

[7] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[8] Y. Himeur, B. Rimal, A. Tiwary, and A. Amira, "Using artificial intelligence and data fusion for environmental monitoring: A review and future perspectives," *Inf. Fusion*, vols. 86–87, pp. 44–75, Oct. 2022.

[9] C. M. Badgujar, A. Poulose, and H. Gan, "Agricultural object detection with you only look once (YOLO) algorithm: A bibliometric and systematic literature review," *Comput. Electron. Agricult.*, vol. 223, Aug. 2024, Art. no. 109090.

[10] M. Ariza-Sentís, S. Vélez, R. Martínez-Peña, H. Baja, and J. Valente, "Object detection and tracking in precision farming: A systematic review," *Comput. Electron. Agricult.*, vol. 219, Apr. 2024, Art. no. 108757.

[11] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, Dec. 2015, pp. 91–99.

[12] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[13] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[14] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[15] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9577–9586.

[16] L. Karlinsky et al., "RepMet: Representative-based metric learning for classification and few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5197–5206.

[17] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[18] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Apr. 2018, pp. 1–7.

[19] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[20] A. Azeem, Z. Li, A. Siddique, Y. Zhang, and S. Zhou, "Unified multimodal fusion transformer for few shot object detection for remote sensing images," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102508.

[21] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.

[22] S. Zhang et al., "Text semantic fusion relation graph reasoning for few-shot object detection on remote sensing images," *Remote Sens.*, vol. 15, no. 5, p. 1187, Feb. 2023.

[23] X. Lu et al., "Few-shot incremental object detection in aerial imagery via dual-frequency prompt," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624017.

[24] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[25] W. Li, J. Zhou, X. Li, Y. Cao, G. Jin, and X. Zhang, "InfRS: Incremental few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644314.

[26] N. Dong, Y. Zhang, M. Ding, and G. H. Lee, "Incremental-DETR: Incremental few-shot object detection via self-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 543–551.

[27] F. Zhang, Y. Shi, Z. Xiong, and X. Xiang Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.

[28] J. Zhang, Z. Hong, X. Chen, and Y. Li, "Few-shot object detection for remote sensing imagery using segmentation assistance and triplet head," *Remote Sens.*, vol. 16, no. 19, p. 3630, Sep. 2024.

[29] W. Guan et al., "Efficient meta-learning enabled lightweight multiscale few-shot object detection in remote sensing images," 2024, *arXiv:2404.18426*.

[30] C. Hao, Z. Li, Y. Zhang, W. Chen, and Y. Zou, "Infrared small target detection based on adaptive size estimation by multidirectional gradient filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5007915.

[31] Y. Li, Z. Li, Z. Guo, A. Siddique, Y. Liu, and K. Yu, "Infrared small target detection based on adaptive region growing algorithm with iterative threshold analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5003715.

[32] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[33] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[34] H.-X. Yu, Y. Tian, Q. Ye, and Y. Liu, "Spatial transform decoupling for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 1–8.

[35] L. Xu, J. Yu, and D. Ming, "HODet: A new detector for arbitrary-oriented rectangular object in optical remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2918–2926, 2024.

[36] Y. Yao, G. Cheng, C. Lang, X. Yuan, X. Xie, and J. Han, "Hierarchical mask prompting and robust integrated regression for oriented object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13071–13084, Dec. 2024.

[37] M. Yuan, X. Shi, N. Wang, Y. Wang, and X. Wei, "Improving RGB-infrared object detection with cascade alignment-guided transformer," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102246.

[38] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[39] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2021.

[40] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.

[41] J. Zhou, W. Li, Y. Cao, H. Cai, and X. Li, "Few-shot oriented object detection with memorable contrastive learning in remote sensing images," 2024, *arXiv:2403.13375*.

[42] J. Wu, C. Qin, and G. Feng, "SMDC-net: Saliency-guided multi-head distribution calibration network for few-shot object detection on remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

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[43] B. Yan, C. Lang, G. Cheng, and J. Han, "Understanding negative proposals in generic few-shot object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5818–5829, Jul. 2024.
[44] Y. Yu, S.-B. Chen, L.-L. Huang, J. Tang, and B. Luo, "Few-shot object detection in remote sensing images with multiscale spatial selective attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[45] J. Wu, C. Lang, G. Cheng, X. Xie, and J. Han, "Retentive compensation and personality filtering for few-shot remote sensing object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5805–5817, Jul. 2024.
[46] X. Bou, G. Facciolo, R. Grompone von Gioi, J.-M. Morel, and T. Ehret, "Exploring robust features for few-shot object detection in satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 430–439.
[47] E. Tulving, *Episodic and Semantic Memory*. New York, NY, USA: Academic, 1972.
[48] E. Tulving, *Elements of Episodic Memory*. London, U.K.: Oxford Univ. Press, 1983.
[49] E. Tulving et al., "Episodic memory: From mind to brain," *Annu. Rev. Psychol.*, vol. 53, no. 1, pp. 1–25, 2002.
[50] G. Zhu, Z. Lin, G. Yang, and C. Zhang, "Episodic reinforcement learning with associative memory," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Apr. 2020, pp. 1–17.
[51] S. J. Gershman and N. D. Daw, "Reinforcement learning and episodic memory in humans and animals: An integrative framework," *Annu. Rev. Psychol.*, vol. 68, no. 1, pp. 101–128, Jan. 2017.
[52] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 6467–6476.
[53] A. Chaudhry et al., "On tiny episodic memories in continual learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jan. 2019, pp. 2621–2631.
[54] A. Santoro, S. Bartunov, M. Botvinick, D. Wierstra, and T. Lillicrap, "Meta-learning with memory-augmented neural networks," in *Proc. Int. Conf. Mach. Learn.*, Jun. 2016, pp. 1842–1850.
[55] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one-shot learning," in *Proc. NIPS*, 2016, pp. 3630–3638.
[56] T. Munkhdalai and H. Yu, "Meta networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 2554–2563.
[57] T. Ramalho and M. Garnelo, "Adaptive posterior learning: Few-shot learning with a surprise-based memory module," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2019, pp. 1–14.
[58] Y. Du, J. Shen, X. Zhen, and C. G. M. Snoek, "EMO: Episodic memory optimization for few-shot meta-learning," in *Proc. CoLLAs*, Jan. 2023, pp. 1–6.
[59] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[60] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. North Amer. Chapter Assoc. Comput. Linguistics*, Jan. 2018, pp. 1–12.
[61] X. Xu, C. Wu, S. Rosenman, V. Lal, W. Che, and N. Duan, "BridgeTower: Building bridges between encoders in vision-language representation learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 10637–10647.
[62] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[63] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.
[64] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2022.
[65] L. Qiao, Y. Zhao, Z. Li, X. Qiu, J. Wu, and C. Zhang, "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8681–8690.
[66] D. A. Ganea, B. Boom, and R. Poppe, "Incremental few-shot instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1185–1194.
[67] S. Wolf, J. Meier, L. Sommer, and J. Beyerer, "Double head predictor based few-shot object detection for aerial imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 721–731.



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